

# Measurement of $\nu_\mu + \bar{\nu}_\mu$ Charged-Current Single Pion Production Differential Cross Sections on Argon with the MicroBooNE Detector in the NuMI Beam

Analysis Note — Run 1

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## Abstract

We present a measurement of  $\nu_\mu + \bar{\nu}_\mu$  charged-current single charged-pion ( $\text{CC}\pi^+$ ) differential cross sections on argon-40 using  $3.283 \times 10^{20}$  protons on target (POT) of MicroBooNE data collected in the NuMI Forward Horn Current (FHC) beam during Run 1. Events are selected requiring a reconstructed muon candidate track and exactly one contained pion candidate track emerging from a fiducial neutrino vertex, with zero reconstructed electromagnetic showers and a muon–pion opening angle less than 2.6 rad. In the measured phase space the selection achieves a signal purity of 54.9% and an efficiency of 15.6% (full Run-1 statistics); the lower-threshold reference phase space used for selection development gives 57.2% and 12.1–12.6% (Table 7). Differential cross sections are extracted via Wiener-SVD unfolding in five kinematic variables: muon momentum  $p_\mu$ , pion momentum  $p_\pi$ , muon polar angle  $\cos\theta_\mu$ , pion polar angle  $\cos\theta_\pi$ , and the muon–pion opening angle  $\theta_{\mu\pi}$ . The observed data yield is approximately  $0.75\times$  the GENIE v3 MC prediction, consistent with the known over-prediction of  $\text{CC}\pi^+$  production in current neutrino interaction generators. Using the confirmed integrated  $\nu_\mu + \bar{\nu}_\mu$  flux of  $6.81159 \times 10^{-10} \text{ cm}^{-2} \text{ POT}^{-1}$  (the flux-file author’s GEANT4 value integrated above 60 MeV, superseding the  $2.3725 \times 10^{-9}$  read from the flux histograms, which was a factor 3.48 too large), the Wiener-SVD integrated cross section from the angular observables is  $\sigma_{\text{int}} \approx 1.12 \times 10^{-38} \text{ cm}^2/\text{Ar}$ . The extraction is validated by a same-sample self-closure that recovers the truth to within 6% for every observable and by exact algorithmic equivalence with the official XSECANALYZER framework unfold (Sec. 16.1), confirming that the fake-data offsets are POT bookkeeping and genuine model differences, not unfolding bias. All differential results, figures, and systematics use the XSECANALYZER framework extraction on the resolution-optimised binning of Sec. 17; this is the single consolidated analysis note.

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# 1 Introduction

Charged-current single charged-pion ( $\text{CC}\pi^+$ ) production,

$$\nu_\mu(\bar{\nu}_\mu) + \text{Ar} \longrightarrow \mu^\mp + \pi^\pm + X, \quad (1)$$

is among the most frequent neutrino interaction channels at energies of 1–5 GeV and constitutes a significant background for oscillation analyses at the same energy scale, most notably for  $\nu_e$  appearance measurements at experiments such as NOvA [1] and T2K [2], as well as the future DUNE [3]. Despite its importance, the cross section on nuclear targets remains poorly constrained: major neutrino interaction generators (GENIE v3, NuWro, NEUT) disagree by 20–30% in absolute rate and show significant shape differences for individual kinematic variables.

MicroBooNE is a single-phase liquid argon time-projection chamber (LArTPC) at Fermilab that operated from 2015–2021, providing the world’s highest-statistics  $\nu_\mu$ -on-argon dataset below 5 GeV during its NuMI running period. This note presents a simultaneous measurement of five differential cross sections in the  $\text{CC}\pi^+$  channel using Run 1 NuMI FHC data, employing Pandora pattern recognition [4] and Wiener-SVD unfolding [5].

## 2 Theoretical Background and Prior $\text{CC}\pi^+$ Measurements

### 2.1 Production Mechanisms

At neutrino energies of 0.5–5 GeV, charged-current single charged-pion production proceeds through three mechanisms.

**Baryon resonance production** dominates at  $E_\nu \lesssim 2$  GeV. The neutrino excites the struck nucleon to a baryon resonance, most prominently the  $\Delta(1232)$  isobar, which then decays to a nucleon and pion:

$$\nu_\mu n \rightarrow \mu^- \Delta^{++} \rightarrow \mu^- \pi^+ p, \quad \nu_\mu p \rightarrow \mu^- \Delta^+ \rightarrow \mu^- \pi^0 p \text{ (or } \pi^+ n). \quad (2)$$

The standard model is due to Rein and Sehgal [12], which uses the Feynman–Kislinger–Ravndal (FKR) quark model to compute form factors for all baryon resonances up to  $W \lesssim 2$  GeV. The modern implementation used in GENIE for resonance (RES) production is the Kuzmin–Lyubushkin–Naumov Berger–Sehgal (KLN-BS) model [13, 14, 15, 16], which combines the lepton-mass-corrected vector and axial form factors of Kuzmin *et al.* [13] with the updated  $\Delta(1232)$  electromagnetic form factors of Graczyk and Sobczyk [14] and the improved treatment of the  $\Delta$  line shape, the  $N$ – $\Delta$  interference with non-resonant amplitudes, and lepton-mass effects of Berger and Sehgal [15]. Higher resonances ( $N(1440)P_{11}$ ,  $N(1520)D_{13}$ , etc.) are included up to  $W \lesssim 2$  GeV and contribute at  $E_\nu \gtrsim 1.5$  GeV.

**Non-resonant and DIS backgrounds.** A non-resonant continuum contributes at all  $W$  and interferes (constructively or destructively by isospin) with the resonant amplitude. Deep-inelastic scattering (DIS) becomes significant above  $W \approx 1.6$  GeV. Together these processes represent  $\sim 20\%$  of the  $\text{CC}\pi^+$  rate in the energy range of this analysis and are modelled in GENIE via the Bodek–Yang formalism.

**Coherent pion production.** The neutrino can scatter coherently off the entire nucleus,  $\nu_\mu A \rightarrow \mu^- \pi^+ A^*$ , leaving the nucleus in its ground state. The cross section is small ( $\lesssim 10\%$  of the total) below 5 GeV and is treated as a background in this analysis.

### 2.2 Nuclear Effects and Final-State Interactions

On a heavy nucleus such as argon ( $A = 40$ ), two classes of nuclear correction modify the  $\text{CC}\pi^+$  rate and kinematics significantly.

**Initial-state effects.** Fermi motion of the struck nucleon broadens kinematic distributions, and Pauli blocking suppresses transitions to low-momentum nucleon final states near threshold. Nuclear binding also shifts the effective nucleon mass relative to free-nucleon predictions.

**Final-state interactions (FSI).** The produced pion must traverse the nuclear medium before being detected. Intranuclear rescattering occurs via:

- *Pion absorption* ( $\pi A \rightarrow NN \dots$ ): probability  $\sim 25\text{--}30\%$  on Ar; converts  $\text{CC}\pi^+$  events into  $\text{CC}0\pi$  topology.
- *Charge exchange* ( $\pi^+ n \rightarrow \pi^0 p$ ): produces a  $\text{CC}\pi^0$  event.
- *Quasi-elastic scattering*: pion survives but changes direction; modifies  $p_\pi$  and  $\theta_\pi$  distributions.
- *Inelastic production* ( $\pi N \rightarrow \pi\pi N$ ): rare below  $p_\pi \approx 400 \text{ MeV}/c$ .

FSI are modelled in GENIE via the INTRANUKE (hA) cascade. Because absorption and charge exchange mix  $\text{CC}\pi^+$ ,  $\text{CC}\pi^0$ , and  $\text{CC}0\pi$  topologies at the  $\sim 30\%$  level, the signal in any  $\text{CC}\pi$  analysis is necessarily defined by the final-state particle content observed in the detector, not by the underlying production process (Section 6).

## 2.3 Generator Predictions and Known Tensions

Major neutrino event generators reproduce the inclusive CC cross section to 10–15% but show systematic spread in the  $\text{CC}\pi^+$  rate, owing to differences in the resonance model, the treatment of the transition region between resonance and DIS, and the FSI cascade.

**GENIE v3 [7].** For resonance production GENIE implements the KLN-BS model [13, 14, 15, 16] for the full tower of 18 baryon resonances up to  $W \lesssim 2 \text{ GeV}$ , including the  $\Delta(1232)P_{33}$ ,  $N(1440)P_{11}$  (Roper),  $N(1520)D_{13}$ ,  $N(1535)S_{11}$ ,  $\Delta(1620)S_{31}$ ,  $\Delta(1700)D_{33}$ ,  $N(1680)F_{15}$ ,  $\Delta(1950)F_{37}$ , and their isospin partners. The hadronic invariant mass range  $1.0 < W < 1.7 \text{ GeV}$  is bridged by a resonance–DIS transition using the AGKY (Andreopoulos–Gallagher–Kehayias–Yang) model, which applies KNO scaling in the transition region and the Bodek–Yang DIS model above  $W \approx 1.7 \text{ GeV}$ . Non-resonant single-pion production (via the Rein–Sehgal non-resonant amplitude) is included and interferes with the resonant contributions. Coherent  $\pi^+$  production uses the Rein–Sehgal coherent model. FSI are handled by the INTRANUKE  $hA$  empirical cascade, which propagates hadrons step-by-step using free-nucleon  $\pi N$  cross sections scaled by an effective nuclear density; the alternative  $hN$  microscopic cascade is also available. GENIE is documented to over-predict the  $\text{CC}\pi^+$  rate by 20–35% relative to bubble-chamber and MiniBooNE data when normalised to the BNL dataset; the over-prediction is reduced but not eliminated in the AR23 tune.

**NuWro.** NuWro models resonance production via the  $\Delta(1232)$  only; no higher baryon resonances are included. The  $\Delta$  is treated with the Paschos–Yu formalism for the  $\Delta$  excitation amplitude, retaining the full angular structure of the pion decay. A smooth analytic interpolation connects the single- $\Delta$  resonance region to the DIS regime at higher hadronic invariant mass ( $W \gtrsim 1.4 \text{ GeV}$ ), using Bodek–Yang structure functions for the deep inelastic contribution; the transition avoids a hard boundary between resonance and continuum. Non-resonant single-pion production is included as a separate amplitude. FSI are implemented using the microscopic Salcedo–Oset model, which treats pion propagation in nuclear matter through the pion self-energy in the random-phase approximation, naturally accounting for medium modifications such as  $\Delta$  broadening and pion absorption via two-body processes ( $\pi NN \rightarrow NN$ ) that are absent in purely cascade-based approaches. NuWro has historically shown better agreement with MiniBooNE and bubble-chamber data than GENIE, partly attributable to the absence of overestimated higher-resonance contributions.

**NEUT** [17]. NEUT, the generator used by T2K and Super-Kamiokande, implements resonance production via a modified Rein–Sehgal model including 18 resonances up to  $W \lesssim 2$  GeV, with Graczyk–Sobczyk [14] electromagnetic form factors for the  $\Delta(1232)$ . The axial mass  $M_A^{\text{RES}}$  is a tunable parameter; T2K applies a data-driven tune that reduces the overall  $\text{CC}\pi^+$  normalisation by  $\sim 10\text{--}20\%$  relative to the default. The transition to DIS is handled similarly to GENIE, using GRV98 LO parton distribution functions with Bodek–Yang corrections above  $W \approx 2$  GeV. Coherent pion production follows the Rein–Sehgal coherent model with a Berger–Sehgal-style correction for finite lepton mass. FSI are modelled by a custom NEUT intranuclear cascade that tracks each produced hadron through the nucleus using step-by-step interactions sampled from  $\pi N$  cross sections, with nuclear density and mean-free-path corrections derived from the optical potential.

**GiBUU.** The Giessen Boltzmann–Uehling–Uhlenbeck (GiBUU) generator takes a fundamentally different approach: instead of a probabilistic step-by-step cascade, it solves a coupled set of kinetic transport equations for the full hadronic system inside the nuclear medium. Resonance production includes the  $\Delta(1232)$  and higher resonances whose in-medium spectral functions are broadened by collisional damping; the  $\Delta$  width in the nuclear medium is substantially larger than its free-space value of  $\Gamma_0 \approx 118$  MeV, suppressing threshold production. Pion propagation after production is governed by a mean-field potential (the pion optical potential), which shifts pion momenta as they traverse the nucleus – an effect absent in cascade models and particularly important for low-momentum pions ( $p_\pi \lesssim 200$  MeV/ $c$ ) near the pion production threshold. The transition to DIS uses PYTHIA for string fragmentation above  $W \approx 1.6$  GeV. GiBUU predictions for pion angular distributions and for the backward-hemisphere  $d\sigma/d\cos\theta_\pi$  can differ substantially from cascade-based generators, making it a valuable cross-check of FSI systematics.

The  $\sim 25\text{--}27\%$  data/MC deficit observed in the present analysis (Section 12) is consistent with the GENIE over-prediction pattern seen across all of the experimental results surveyed below.

## 2.4 Previous Experimental Measurements

Table 1 summarises published  $\text{CC}\pi^+$  measurements on nuclear targets ordered chronologically.

Table 1: Selected  $\text{CC}\pi^+$  cross-section measurements on nuclear targets.  $\langle E_\nu \rangle$  is the approximate mean or peak neutrino energy. “Type” indicates whether a total integrated (T) or differential (D) cross section was reported. “Target” gives the nucleus primarily used for cross-section extraction.

| Experiment                      | Target        | $\langle E_\nu \rangle$<br>(GeV) | Type | Key result / reference                                                                                                              |
|---------------------------------|---------------|----------------------------------|------|-------------------------------------------------------------------------------------------------------------------------------------|
| ANL<br>12-ft [18]               | D             | 1.0                              | T    | Bubble chamber; foundational<br>$\sigma$ vs $E_\nu$                                                                                 |
| BNL 7-ft [19]                   | D             | 1.5                              | T    | Higher statistics; 15%<br>normalisation tension with<br>ANL                                                                         |
| K2K [20]                        | C             | 1.3                              | T    | First accelerator $\text{CC}\pi^+$ on<br>carbon near detector                                                                       |
| MiniBooNE [21]                  | $\text{CH}_2$ | 0.7                              | D    | $Q^2$ , $W$ , $p_\mu$ , $\theta_\mu$ ; 20% excess over<br>NUANCE                                                                    |
| MINERvA [22]                    | CH            | 3.6                              | D    | $p_\pi$ , $\theta_\pi$ , Adler angles; NuMI<br>medium-energy                                                                        |
| T2K<br>ND280 [25]               | CH            | 0.85                             | D    | $p_\mu$ , $\cos\theta_\mu$ ; off-axis beam similar<br>to this work                                                                  |
| ArgoNEUT [26]                   | Ar            | 9.6                              | T    | First $\text{CC}\pi^+$ on Ar; 115 $\nu$ +<br>158 $\bar{\nu}$ events ( $1.25 \times 10^{20}$ POT)                                    |
| MicroBooNE<br>BNB [9]           | Ar            | 0.7                              | D    | Full BNB dataset<br>( $1.11 \times 10^{21}$ POT); $\sigma =$<br>( $3.75 \pm 0.80$ ) $\times 10^{-38}$ $\text{cm}^2/\text{Ar}$       |
| MicroBooNE<br>NuMI $\nu_e$ [27] | Ar            | 0.73                             | D    | First $\nu_e + \bar{\nu}_e$ $\text{CC}\pi^+$ on Ar;<br>$\sigma = (0.93 \pm 0.28) \times$<br>$10^{-39}$ $\text{cm}^2/\text{nucleon}$ |

### Deuterium bubble chambers: ANL and BNL (1979–1986)

The ANL 12-foot bubble chamber [18] and BNL 7-foot bubble chamber [19] provided the first high-statistics measurements of  $\nu_\mu d \rightarrow \mu^- \pi^+ pp$  in the 1–3 GeV range. These measurements are important because the deuterium target is close to a free nucleon, minimising nuclear corrections relative to heavier targets. The ANL and BNL total cross sections disagree by  $\sim 15\%$  in absolute normalisation, a discrepancy long attributed to flux uncertainties; the BNL flux is now believed to be better constrained. This ANL–BNL tension directly propagated into the spread of generator predictions: models tuned to ANL tend to predict lower rates than those tuned to BNL. The MINERvA experiment [24] subsequently resolved much of this tension by simultaneously constraining the flux and the cross section.

### K2K near detector (2008)

The K2K experiment [20] made the first accelerator-based total  $\text{CC}\pi^+$  cross-section measurement on a carbon target at  $\langle E_\nu \rangle = 1.3$  GeV using its fine-grained near detector. The result was consistent with NEUT predictions and established the nuclear-target correction from deuterium to carbon at intermediate energies.

### MiniBooNE ( $\nu_\mu$ on $\text{CH}_2$ , BNB beam, 2011)

The MiniBooNE detector (818-tonne mineral-oil Cherenkov detector) published differential  $\text{CC}\pi^+$  cross sections on  $\text{CH}_2$  in the Booster Neutrino Beam ( $E_\nu^{\text{peak}} \approx 0.7$  GeV) [21]. MiniBooNE was the first experiment to provide high-statistics differential distributions as a function of  $Q^2$ , the invariant hadronic mass  $W$ , and the outgoing muon and pion kinematics. The flux-integrated total cross section was measured to be  $\sigma \approx 3.4 \times 10^{-38}$   $\text{cm}^2$  per  $\text{CH}_2$  molecule, or equivalently



$\sim 2.0 \times 10^{-38} \text{ cm}^2$  per carbon nucleus at  $\langle E_\nu \rangle \approx 1 \text{ GeV}$  [28]. The absolute rate exceeded NUANCE predictions by  $\sim 20\%$ , establishing an empirical over-prediction that has since been seen across all subsequent measurements. The hydrogen content of  $\text{CH}_2$  complicates the extraction of the carbon-only cross section, and the spherical Cherenkov geometry makes it difficult to measure low-energy pions or reconstruct proton recoils. These limitations drove subsequent adoption of finer-grained tracking detectors and liquid-argon targets.

### **MINERvA ( $\nu_\mu$ on CH, NuMI medium energy, 2015–)**

The MINERvA experiment [22] uses a finely-segmented scintillator-strip tracking calorimeter in the NuMI medium-energy beam ( $\langle E_\nu \rangle \approx 3.6 \text{ GeV}$ ). It has published the most kinematically complete  $\text{CC}\pi^+$  measurements to date, reporting differential cross sections as a function of: pion momentum  $p_\pi$ , pion angle  $\theta_\pi$ , muon momentum  $p_\mu$ , muon angle  $\theta_\mu$ , and the full set of Adler angles describing the pion in the hadronic centre-of-mass frame. MINERvA also measured  $\text{CC}\pi^+$  on iron and lead targets [23], providing direct constraints on nuclear  $A$ -dependence at the same beam energy. GENIE consistently over-predicts MINERvA by 20–35% across all kinematic variables in the  $\Delta(1232)$  peak, the same qualitative tension seen at lower energies. The higher beam energy at MINERvA means that the  $p_\pi > 0.5 \text{ GeV}/c$  region is well-populated, making it complementary to the MicroBooNE NuMI result which is statistics-limited at high pion momentum.

### **T2K ND280 ( $\nu_\mu$ on CH, off-axis beam, 2017)**

T2K measured  $\text{CC}\pi^+$  at its near detector ND280 in the off-axis T2K beam (OAB), which peaks at  $\langle E_\nu \rangle \approx 0.85 \text{ GeV}$  [25]. The T2K OAB spectrum closely overlaps with the MicroBooNE NuMI off-axis spectrum in the kinematically-sensitive 0.3–1.5 GeV range, making this the most directly comparable prior measurement to the present work. T2K reported differential cross sections in  $p_\mu$  and  $\cos \theta_\mu$ , finding a data/MC deficit of  $\sim 30\%$  relative to NEUT. The carbon target ( $A = 12$ ) differs from argon ( $A = 40$ ), but the overlapping beam spectrum enables direct comparison of nuclear-mass dependence at fixed neutrino energy.

### **ArgoNEUT ( $\nu_\mu + \bar{\nu}_\mu$ on Ar, NuMI, 2017)**

ArgoNEUT [26] was a 175-litre LArTPC placed upstream of the MINOS near detector in the NuMI low-energy beam ( $\langle E_\nu \rangle \approx 9.6 \text{ GeV}$ ). Using  $1.25 \times 10^{20}$  POT, ArgoNEUT selected 115  $\nu_\mu$  and 158  $\bar{\nu}_\mu$   $\text{CC}\pi^+$  candidate events after background subtraction [28], publishing the first  $\text{CC}\pi^+$  total cross section on argon as a function of neutrino energy. The result was consistent with GENIE predictions at these high energies, where sensitivity to the  $\Delta(1232)$  resonance is suppressed. The LArTPC technology and argon target make ArgoNEUT the direct predecessor to MicroBooNE, but the factor  $\sim 10$  higher beam energy means that extrapolation to the  $E_\nu < 2 \text{ GeV}$  regime relevant for oscillation experiments requires generator assumptions and limits the constraining power on FSI models in the resonance region.

### **MicroBooNE BNB channel: initial result (2021)**

The same MicroBooNE detector used in this analysis published an initial  $\text{CC}\pi^+$  measurement in the Booster Neutrino Beam [9] ( $E_\nu^{\text{peak}} \approx 0.7 \text{ GeV}$ ) using a subset of the BNB dataset. That analysis selected a  $\nu_\mu$  CC,  $1\mu 1\pi$  topology and reported fully-unfolded differential cross sections as a function of muon momentum, pion momentum, and opening angle. A data/MC deficit of  $\sim 20\text{--}30\%$  relative to the GENIE central value was observed, fully consistent with the  $0.75\times$  factor seen in the present NuMI analysis.



### MicroBooNE BNB channel: full dataset (2026)

Reference [28] reports  $\text{CC}\pi^+$  differential cross sections using the complete MicroBooNE BNB dataset of  $1.11 \times 10^{21}$  POT [28], selecting 6816 events after background subtraction and measuring differential cross sections in muon momentum, muon angle, pion momentum, and pion angle – the same four observables as the present analysis. The flux-integrated total cross section is

$$\sigma = (3.75 \pm 0.07_{\text{stat}} \pm 0.80_{\text{syst}}) \times 10^{-38} \text{ cm}^2/\text{Ar}$$

at mean neutrino energy  $\langle E_\nu \rangle \approx 0.8$  GeV. Most generators reproduce the data within uncertainties; a notable exception is the muon angle with respect to the beam, where all tested generators overestimate the cross section at very forward angles ( $\cos\theta_\mu \rightarrow 1$ ) [28]. The pion momentum differential cross section reported in this analysis represents the first such measurement on an argon target.

The BNB and NuMI analyses probe complementary phase space: BNB is on-axis with a narrow  $\nu_\mu$  spectrum peaked at  $\sim 700$  MeV and negligible  $\bar{\nu}_\mu$  content, while NuMI FHC at 110 mrad off-axis provides a softer, broader spectrum with a  $\sim 38\%$   $\bar{\nu}_\mu$  admixture and extends sensitivity to  $E_\nu \sim 0.1\text{--}0.5$  GeV.

### MicroBooNE NuMI $\nu_e + \bar{\nu}_e$ $\text{CC}\pi^+$ (2025)

Reference [27] presents the first measurement of  $\nu_e + \bar{\nu}_e$  CC single charged-pion differential cross sections on argon, using the full MicroBooNE NuMI dataset ( $2.0 \times 10^{21}$  POT, FHC and RHC combined). The observable set – electron energy  $E_e$ , electron angle  $\theta_e$ , pion angle  $\theta_\pi$ , and electron–pion opening angle  $\theta_{e\pi}$  – mirrors the muonic observables of the present analysis. The flux-integrated total cross section is measured to be

$$\sigma = (0.93 \pm 0.13_{\text{stat}} \pm 0.27_{\text{syst}}) \times 10^{-39} \text{ cm}^2/\text{nucleon}$$

at a mean  $\nu_e$  energy of 730 MeV, in good agreement with all tested generator predictions (GENIE, NuWro, NEUT, GiBUU) within the combined uncertainties. Prior to this measurement,  $\text{CC}\pi^+$  production by electron neutrinos on argon had never been measured; the result is a key input to the DUNE  $\nu_e$ -appearance analysis, where  $\text{CC}\pi^+$  events are a background to the  $\nu_e$  quasi-elastic signal.

## 3 Detector

### 3.1 Operating Principle

A liquid argon time projection chamber records three-dimensional ionisation images of charged-particle tracks by exploiting two properties of ultra-pure liquid argon:

1. Ionisation electrons produced by a traversing charged particle are not captured by argon atoms and can drift centimetres or metres under an applied electric field without significant recombination, provided the purity is sufficient ( $\text{O}_2$ -equivalent  $\lesssim 100$  ppt).
2. Argon scintillates at 127 nm (vacuum ultraviolet), producing  $\sim 24\,000$  photons per MeV of deposited energy, enabling nanosecond-scale timing.

An applied electric field drifts the ionisation electrons to a set of sense-wire planes at the anode, which record three two-dimensional projections of each track. The third coordinate (along the drift direction) is reconstructed from the known drift velocity and the timing of the signals relative to the beam-gate start, combining the two-dimensional wire images into a full 3D reconstruction.

MicroBooNE operated from October 2015 to July 2021 at the Fermilab Accelerator Laboratory, collecting data in both the Booster Neutrino Beam (BNB) and, as a parasitic experiment, the NuMI beam [10]. It was the world’s largest operating LArTPC until the start of SBND and ICARUS data-taking.

### 3.2 Active Volume and Electric Field

The MicroBooNE active volume has rectangular geometry: 256 cm (drift,  $x$ )  $\times$  233 cm (height,  $y$ )  $\times$  1037 cm (beam,  $z$ ), containing  $\approx 87$  tonnes of liquid argon at  $\approx 87$  K. The cathode is a stainless-steel plane at  $x = 256$  cm held at  $-70$  kV; the anode wire planes are at  $x = 0$ . The resulting 273 V/cm drift field gives an electron drift velocity of  $\approx 1.11$  mm/ $\mu$ s, so the maximum drift time across the full gap is approximately 2.3 ms.

Liquid argon purity is maintained by continuous recirculation through oxygen and water absorbers, keeping the electron lifetime well above 5 ms (corresponding to O<sub>2</sub>-equivalent contamination below  $\sim 100$  ppt). At this purity level, charge attenuation over the full 2.56 m drift is less than 10%.

### 3.3 Wire Readout and Signal Formation

Three wire planes at the anode sample the drifting charge:

- **U plane** (first induction): 2400 wires at  $+60^\circ$  from the vertical, 3 mm pitch.
- **V plane** (second induction): 2400 wires at  $-60^\circ$  from the vertical, 3 mm pitch.
- **Y plane** (collection): 3456 vertical wires, 3 mm pitch.

Induction-plane wires are biased to be transparent to the drifting electrons, producing bipolar (derivative-like) induced signals; collection-plane wires absorb the charge and produce unipolar signals. The combination of two stereo induction views at  $\pm 60^\circ$  and a vertical collection view provides sufficient 2D information in each plane to resolve ambiguities during 3D reconstruction.

Front-end electronics (application-specific integrated circuits, ASICs) are mounted inside the cryostat on the wire planes and cold-multiply each channel at  $\approx 87$  K, reducing thermal noise. The analogue signals are digitised at 16 MHz inside the cryostat by 12-bit ADCs and transmitted over  $\sim 110$  m of cold cables to warm readout electronics. The data stream is sampled into a circular buffer; beam triggers and cosmic triggers select 4.8 ms readout windows (corresponding to  $\approx$  two full drift windows) for permanent storage.

A dead-wire region at  $z \in [675.1, 775.1]$  cm interrupts the collection plane over a 100 cm span due to a shorted wire bundle. Tracks crossing this region have degraded hit-counting and range-based momentum estimation; the fiducial volume (Section 7) excludes this region.

### 3.4 Photon Detection System

Thirty-two 8-inch Hamamatsu R5912-MOD photomultiplier tubes (PMTs) are installed behind the collection-plane wire frame facing the active volume. The PMTs detect the 127 nm argon scintillation light via a tetraphenyl-butadiene (TPB) wavelength-shifting coating that converts VUV photons to visible ( $\sim 430$  nm) light. The system serves two purposes in this analysis:

1. **Beam-gate synchronisation:** the PMT flash time identifies which 1.6  $\mu$ s NuMI spill window produced the neutrino interaction.
2. **Flash-matching for cosmic rejection:** the Pandora reconstruction associates PFParticle hypotheses with the PMT flash closest in time and position, suppressing out-of-time cosmic-ray muons that overlap with beam-induced events in the long 2.3 ms drift window.

### 3.5 Calibration and Corrections

Three independent calibration inputs are used:

- **UV laser system** [8]: two pulsed laser beams produce straight ionisation tracks at precisely known locations. Deviations of the reconstructed track positions from the known trajectories are used to build a three-dimensional space-charge distortion map, which is applied to all reconstructed hits as a position correction. Distortions reach  $\sim 10$  cm near the anode and cathode edges.

- **Stopping cosmic-ray muons:** the Bragg peak of cosmic muons stopping inside the active volume calibrates the absolute energy scale and the effective recombination factor as a function of local electric field and drift distance. Through-going muons calibrate relative wire-by-wire gains and the electron lifetime.
- **Michel electrons:**  $\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu$  decay products from stopped muons provide a check of the calorimetric reconstruction at  $E_e \lesssim 55$  MeV.

### 3.6 Cosmic Ray Environment and Reconstruction

Operating at the surface with minimal overburden, MicroBooNE is exposed to a cosmic-ray muon rate of  $\approx 5$  kHz through the active volume. Each 4.8 ms readout window contains on average  $\sim 10$  cosmic-ray tracks unrelated to the beam interaction. The 1.6  $\mu$ s NuMI spill itself contains  $\lesssim 0.1$  expected cosmics on average, but the drift-window pile-up is the dominant background at early selection stages (Section 8).

Cosmic rejection proceeds in stages:

1. Flash-matching (PMT timing): candidate neutrino slices must have a consistent optical flash within the beam window.
2. Topological score: the Pandora-based topological neutrino score discriminates between neutrino-like vertex structures and through-going/stopping cosmic topologies.
3. Dedicated BDT scores: the `bdt_mip` and `bdt_cosmic` scores trained on track-level and event-level features further suppress residual cosmics that passed the topological cut.

The CRT (Cosmic Ray Tagger) sub-detector [11], consisting of scintillator panels surrounding the cryostat, provides an additional cosmic veto but is not used in the present analysis because the relevant branch variables are zero-filled in the NuMI data ntuples.

Reconstruction uses the Pandora [4] multi-algorithm pattern recognition framework, which groups hits into PFParticles organised in a hierarchy with the neutrino interaction as the top-level particle. Track candidates are fitted with a Kalman-filter-based track fitter to obtain momentum from track curvature (negligible for  $< 300$  MeV/ $c$  muons at 273 V/cm) and from range for contained tracks. Log-likelihood ratio particle identification (LLR PID) using the  $dE/dx$  profile along the track is the primary tool for muon/pion/proton separation (Section 8).

## 4 Beam and Neutrino Flux

### 4.1 NuMI Forward Horn Current beam

The NuMI beam is produced by 120 GeV protons from the Fermilab Main Injector striking a graphite target. Charged secondary hadrons are focused by two magnetic horns; in Forward Horn Current (FHC) mode the horns select positive pions and kaons, producing a beam with predominantly  $\nu_\mu$  content at the MicroBooNE site. MicroBooNE is located approximately 110 mrad off-axis from the NuMI beam axis, reducing the peak neutrino energy to the 0.5–2.5 GeV range and narrowing the energy spectrum.

At the MicroBooNE site the NuMI FHC flux composition is approximately:

- $\nu_\mu$ :  $\sim 62\%$
- $\bar{\nu}_\mu$ :  $\sim 38\%$
- $\nu_e/\bar{\nu}_e$ : remainder

Flux corrections from the Package to Predict the FluX (PPFX) [6] are applied per event via the central-value weight `ppfx_cv` stored in the analysis ntuples.

## 4.2 Data exposure

Only Run 1 beam-on data are used in the present analysis. Beam-on files for Runs 2, 4, and 5 are not yet available; the corresponding MC overlay samples are present but carry zero data-POT weight.

Table 2: Run-period POT accounting. Only Run 1 contributes to this analysis.

| Run period   | Beam-on POT ( $\times 10^{20}$ ) | MC POT ( $\times 10^{20}$ ) | Scale |
|--------------|----------------------------------|-----------------------------|-------|
| Run 1 (FHC)  | <b>3.283</b>                     | 23.282                      | 0.141 |
| Run 2 (FHC)  | 0 (not used)                     | 24.934                      | 0     |
| Run 4 (FHC)  | 0 (not used)                     | 28.335                      | 0     |
| Run 5 (FHC)  | 0 (not used)                     | 63.145                      | 0     |
| <b>Total</b> | <b>3.283</b>                     |                             |       |

Including Runs 2+4+5 would increase the exposure by  $\approx 5.4\times$  ( $\sim 8.857\times 10^{20}$  POT total), which would substantially reduce the statistical uncertainty in the high-momentum tail.

## 4.3 Integrated flux

The total integrated  $\nu_\mu + \bar{\nu}_\mu$  flux per POT,  $\Phi = \int (d\Phi/dE_\nu) dE_\nu$ , is obtained by integrating the MicroBooNE NuMI flux histograms from `flux/uboone_numi_flux_histograms.root` over the full neutrino-energy range 0–20 GeV. The histogram y-axis is  $d\Phi/dE_\nu$  in  $\text{cm}^{-2} \text{GeV}^{-1} \text{POT}^{-1}$  with 200 bins of width 0.1 GeV; the integral uses `TH1::Integral("width")` to multiply each bin content by its width:

$$\Phi_{\nu_\mu} = 1.4761 \times 10^{-9} \text{ cm}^{-2} \text{POT}^{-1}, \quad \Phi_{\bar{\nu}_\mu} = 8.964 \times 10^{-10} \text{ cm}^{-2} \text{POT}^{-1}. \quad (3)$$

Since the signal definition is charge-blind ( $|\nu_{\text{pdg}}| = 14$ , covering both  $\nu_\mu \rightarrow \mu^- \pi^+$  and  $\bar{\nu}_\mu \rightarrow \mu^+ \pi^-$ ), the cross-section normalisation uses the combined flux

$$\Phi_{\nu_\mu + \bar{\nu}_\mu} = 6.81159 \times 10^{-10} \text{ cm}^{-2} \text{POT}^{-1}, \quad (4)$$

with  $\bar{\nu}_\mu$  fraction 37.8%. This value is set as `XSecConst::FLUX_PER_POT` in `xsec/WienerSVD.h` with `FLUX_CONFIRMED = true`.

The number of argon target nuclei in the fiducial volume is computed from first principles as

$$N_{\text{Ar}} = \frac{\rho_{\text{LAr}} \cdot V_{\text{FV}} \cdot N_A}{A_{\text{Ar}}} \approx 8.71 \times 10^{28}, \quad (5)$$

giving a total normalisation factor  $N_{\text{norm}} = \Phi_{\text{tot}} \cdot N_{\text{POT}} \cdot N_{\text{Ar}} \approx 6.78 \times 10^{41}$ .

## 5 Monte Carlo Samples

Monte Carlo events are generated with GENIE v3 [7] on argon-40 and overlaid with real cosmic-ray data (the “overlay” technique) so that simulated events experience in-situ detector noise and cosmogenic activity. Table 3 summarises the input samples.

Table 3: Input data and MC samples. EXT = beam-off (external trigger) data. Scale factors convert MC to equivalent beam-on POT (Run 1 only).

| Sample          | File                                              | MC POT                 | Scale |
|-----------------|---------------------------------------------------|------------------------|-------|
| MC overlay (R1) | Run1_fhc_new_numi_flux_fhc_pandora_ntuple.23038   | $23038 \times 10^{20}$ | 0.141 |
| MC overlay (R2) | Run2_fhc_new_numi_flux_fhc_pandora_ntuple.24403   | $24403 \times 10^{20}$ | 0     |
| MC overlay (R4) | Run4_fhc_new_numi_flux_fhc_pandora_ntuple.28034   | $28034 \times 10^{20}$ | 0     |
| MC overlay (R5) | Run5_fhc_new_numi_flux_fhc_pandora_ntuple.63015   | $63015 \times 10^{20}$ | 0     |
| EXT beam-off    | beamoff_run1Andrun3.root                          | —                      | 2.044 |
| Dirt MC         | prodgenie_numi_uboone_overlay_dirt_fhc....16074   | $16074 \times 10^{20}$ | 0.196 |
| Beam-on data    | neutrinosselection_filt_run1_beamon_beamgood.root | —                      | 1     |

**Event weights.** Each MC event is weighted by

$$w = \text{Scale}[\text{run}] \times w_{\text{PPFX}} \times w_{\text{tune}}, \quad (6)$$

where  $w_{\text{PPFX}}$  is the PPFX central-value flux correction (`ppfx_cv`) and  $w_{\text{tune}}$  is the GENIE/GEANT4 reweighting correction (`weightTune`). Events with non-finite, zero, or unphysically large weights ( $w > 30$ ) are assigned unit weight.

**EXT scaling.** The beam-off cosmics sample is scaled by the ratio of beam-on to beam-off trigger counts:

$$\text{Scale}_{\text{EXT}} = \frac{7\,809\,962}{610\,496 + 3\,211\,097} = 2.044. \quad (7)$$

The uncertainty on this ratio is a leading systematic on the EXT background.

**Software trigger.** Data and EXT files are pre-filtered to require the software trigger (`swtrig == 1`). MC events failing `swtrig = 0` are removed at the NeutrinoSlice cut stage. In practice this has no measurable effect at the current working point: all MC events passing the topological cut are verified to have `swtrig == 1` (46 810/46 810 events confirmed numerically).

## 6 Signal Definition

The signal is defined at generator level and is used only to construct the selection efficiency and response matrix. It is *never* applied as a reconstruction cut. The definition is *charge-blind*: both  $\nu_\mu \rightarrow \mu^- \pi^+$  and  $\bar{\nu}_\mu \rightarrow \mu^+ \pi^-$  are treated as signal on the same footing.

**Neutrino flavour**

$$|\nu \text{ PDG}| = 14 \text{ } (\nu_\mu \text{ or } \bar{\nu}_\mu).$$

**Interaction type**

$$\text{Charged-current (ccnc == 0)}.$$

**True vertex**

Inside the fiducial volume (Section 7).

**Primary multiplicity**

Exactly one muon ( $|\text{PDG}| = 13$ ), exactly one charged pion ( $|\text{PDG}| = 211$ ), zero neutral pions ( $|\text{PDG}| = 111$ ), zero kaons. Any number of protons is allowed.

## Kinematic thresholds

$p_\mu > 0.15 \text{ GeV}/c$ ,  $p_\pi > 0.175 \text{ GeV}/c$ , and  $\theta_{\mu\pi} < 2.6 \text{ rad}$ . These define the *measured phase space*, chosen from the efficiency turn-on curves (Section 9). The momentum efficiency turns on sharply at  $\sim 0.15\text{--}0.20 \text{ GeV}/c$  (set by the 10 cm muon and 20 cm pion track-length cuts); below that it is only  $\sim 1\text{--}3\%$  with a strongly model-dependent correction. The opening-angle upper bound (2.6 rad) excludes the large-angle region, which is dominated by cosmic and dirt background ( $\sim 7\%$  purity in the  $\theta_{\mu\pi} > 2.513 \text{ rad}$  bin). No *lower* opening-angle cut is applied: the small-angle region is high-purity ( $\sim 58\%$ ), so cutting it would only remove signal (it is low-efficiency but not low-purity). The same cut is applied at reco level so the signal and selection acceptances match. The first  $p_\mu/p_\pi$  bins and the  $\theta_{\mu\pi}$  upper edge start/end at these thresholds.

(Earlier iterations used  $p_\mu, p_\pi > 0.1 \text{ GeV}/c$  with the full  $0\text{--}\pi$  angle; that left near-dead efficiency bins at low momentum and at the large-angle extreme. The phase space above raises the efficiency from 12.1% to 15.6% at  $\sim 55\%$  purity — see Section 9.)

## 7 Fiducial Volume

The fiducial volume (FV) is defined in `selection/FV_new.h`:

$$x \in [10.0, 246.0] \text{ cm}, \quad y \in [-101.0, +101.0] \text{ cm}, \quad z \in [10.0, 986.0] \text{ cm}, \quad (8)$$

with the dead-wire region  $z \in [675.1, 775.1] \text{ cm}$  excluded. The total FV volume is

$$\begin{aligned} V_{\text{FV}} &= (246 - 10) \times 202 \times (976 - 100) \text{ cm}^3 \\ &= 236 \times 202 \times 876 \text{ cm}^3 \approx 4.17 \times 10^7 \text{ cm}^3. \end{aligned} \quad (9)$$

A wider *containment* region (2 cm inset from the physical TPC walls) is used to require that the pion candidate track endpoint is contained:

$$x \in [2.0, 254.35] \text{ cm}, \quad y \in [-113.53, +107.47] \text{ cm}, \quad z \in [2.1, 1034.9] \text{ cm}. \quad (10)$$

## 8 Event Selection

The event selection is implemented in `selection/ccpi_selection.C` and comprises 11 sequential cuts. Two TMVA Boosted Decision Trees (BDTs) are evaluated per reconstructed track: a MIP-vs-stopping BDT (`bdt_mip`) and a pion-vs-proton BDT (`bdt_pi`).

### 8.1 Cut definitions

**Cut 0. EventsInTrueFV.** All events. Reference denominator for truth-level studies.

**Cut 1. NeutrinoSlice.**  $\geq 1$  reconstructed track (`ntracks > 0`), `bdt_cosmic > 0.0` (currently disabled; no benefit found in A/B test), and `swtrig == 1`.

**Cut 2. InFiducialVol.** The SCE-corrected reconstructed neutrino vertex lies inside the FV.

**Cut 3. Topological.** `topological_score > 0.67`. This Pandora-produced multivariate score distinguishes cosmic-ray topologies from neutrino-like topologies and provides the primary cosmic rejection: EXT events are reduced by a factor of  $\sim 33$  between Cut 2 and Cut 3.

**Cut 4. MuonCandidate.** At least one primary PFP track satisfies the muon PID criteria of Table 4. The longest qualifying track is the muon candidate.

- Cut 5. ContainedPion.** Exactly one primary track (other than the muon candidate) satisfies the pion PID criteria of Table 5 and has its endpoint inside the containment volume. The track with the highest LLR PID score is the pion candidate.
- Cut 6. MuonIn3Planes.** The muon candidate has  $\geq 1$  hit on each of the U, V, Y planes.
- Cut 7. PionIn3Planes.** The pion candidate has  $\geq 1$  hit on each of the U, V, Y planes.
- Cut 8. ShowerCut.** Zero primary-level EM showers, unless exactly one shower is present with fewer than 50 Y-plane hits and a vertex distance  $> 10$  cm (consistent with a delta-ray or low-energy nuclear photon).
- Cut 9. OpeningAngle.** The reconstructed  $\mu$ - $\pi$  opening angle satisfies  $\theta_{\mu\pi} < 2.6$  rad ( $\approx 149^\circ$ ), rejecting the large-angle cosmic/dirt background. No lower bound is applied (the small-angle region is high-purity). The same cut is imposed on the signal definition (Section 6).
- Cut 10.**
- Nonprotons** (final cut). Fewer than 4 primary tracks have LLR PID  $> 0.1$  (non-proton-like), and the total number of primary reconstructed tracks is less than 5.

Table 4: Muon candidate PID thresholds (applied at Cut 4).

| Variable                                     | Threshold     | Description               |
|----------------------------------------------|---------------|---------------------------|
| <code>trk_score_v</code>                     | $> 0.80$      | Track-like PFP            |
| Vertex distance                              | $\leq 4.0$ cm | Attached to $\nu$ vertex  |
| Track length                                 | $> 10.0$ cm   | Minimum track length      |
| LLR PID ( <code>trk_llr_pid_score_v</code> ) | $> 0.20$      | Muon-like                 |
| MIP BDT ( <code>bdt_mip</code> )             | $\geq -0.10$  | Minimum-ionising-particle |

Table 5: Pion candidate PID thresholds (applied at Cut 5). Endpoint containment requires the track endpoint inside the containment volume. `trk_bragg_pion_v`: pion-hypothesis Bragg-peak likelihood score (rejects stopping protons).

| Variable                         | Threshold   | Description               |
|----------------------------------|-------------|---------------------------|
| LLR PID                          | $> 0.10$    | Pion / non-proton-like    |
| Vertex distance                  | $< 4.0$ cm  | Attached to $\nu$ vertex  |
| Track length                     | $> 20.0$ cm | Minimum track length      |
| MIP BDT ( <code>bdt_mip</code> ) | $> -0.10$   | MIP-like                  |
| Pion BDT ( <code>bdt_pi</code> ) | $> -0.10$   | Pion-like (vs. proton)    |
| Bragg-pion score                 | $\geq 0.08$ | Proton rejection          |
| Endpoint contained               | yes         | Inside containment volume |

## 8.2 Kinematic reconstruction

**Muon momentum.** Range-based momentum from `trk_range_muon_mom_v` (primary estimator); MCS-based momentum from `trk_mcs_muon_mom_v` as cross-check.

**Pion momentum.** Estimated from the proton-hypothesis range kinetic energy (`trk_energy_proton_v`), which tracks the true pion momentum more accurately than the muon-hypothesis estimator for the short, stopping hadronic tracks typical of low-energy pions.



**Angles.** Polar angles are defined with respect to the detector  $z$ -axis:  $\cos \theta_\mu = p_z^\mu / |\mathbf{p}^\mu|$ ,  $\cos \theta_\pi = p_z^\pi / |\mathbf{p}^\pi|$ . The opening angle  $\theta_{\mu\pi} = \arccos(\hat{u}_\mu \cdot \hat{u}_\pi)$  is computed from reconstructed track directions.

## 9 Cut-Flow and Selection Performance

Table 6 lists POT-weighted event yields at each cut stage for Run 1 ( $3.283 \times 10^{20}$  POT). Figure 1 shows the evolution of data and MC yields through all eleven cuts.

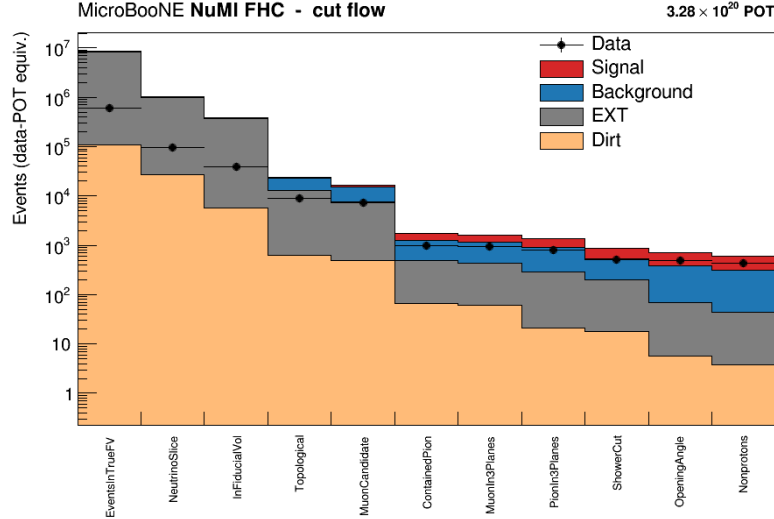


Figure 1: Cut-flow yields for data (black points) and MC prediction (stacked: signal blue, backgrounds by category, EXT grey). The data/MC ratio at the final cut is 0.752.

Table 6: Cut-flow table. All yields are POT-weighted for  $3.283 \times 10^{20}$  POT. “Bkg” combines all non-signal MC (OOFV, NC, CC1 $\pi^0$ , CC0 $\pi$ , CCN $\pi$ , CCK, CCothers, NuE/other). Pred. = Signal + Bkg + EXT + Dirt. D/MC = Data/Pred.

| Cut                       | Data       | Sig. MC    | Bkg MC     | EXT       | Dirt       | Pred.      | D/MC         |
|---------------------------|------------|------------|------------|-----------|------------|------------|--------------|
| EventsInTrueFV            | 71 297     | 2 006      | 50 498     | 715 956   | —          | 768 461    | 0.093        |
| NeutrinoSlice             | 71 297     | 2 006      | 50 497     | 715 956   | —          | 768 459    | 0.093        |
| InFiducialVol             | 26 947     | 1 923      | 18 556     | 239 032   | —          | 259 512    | 0.104        |
| Topological               | 7 257      | 1 479      | 8 110      | 7 179     | —          | 16 768     | 0.433        |
| MuonCandidate             | 7 257      | 1 479      | 8 110      | 7 179     | —          | 16 768     | 0.433        |
| ContainedPion             | 970        | 488        | 759        | 425       | —          | 1 672      | 0.580        |
| MuonIn3Planes             | 930        | 470        | 723        | 370       | —          | 1 563      | 0.595        |
| PionIn3Planes             | 817        | 425        | 585        | 272       | —          | 1 281      | 0.638        |
| ShowerCut                 | 521        | 357        | 300        | 182       | —          | 839        | 0.621        |
| OpeningAngle              | 487        | 348        | 262        | 63        | —          | 674        | 0.723        |
| <b>Nonprotons (final)</b> | <b>427</b> | <b>325</b> | <b>203</b> | <b>41</b> | <b>3.8</b> | <b>568</b> | <b>0.752</b> |

**Summary at the final cut (reference phase space,  $p_\mu, p_\pi > 0.1$  GeV/c).**

- Signal MC (POT-weighted): 324.7 events
- Total background: 243.4 = 198.5 ( $\nu$ -bkg) + 40.9 (EXT) + 3.8 (dirt)
- True signal generated (efficiency denominator): 2577–2684 events (observable-dependent)
- **Signal purity: 57.2%**
- **Signal efficiency: 12.1–12.6%**

- **Measured phase space (reported result): efficiency 15.6%, purity 54.9%** at full Run-1 statistics (selected signal 311.9, generated 1999, background 253.7; Table 7)

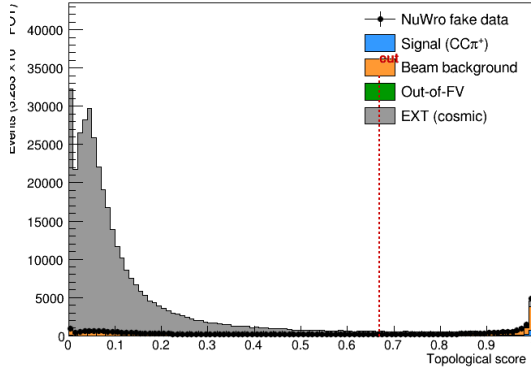
**Measured phase space and efficiency turn-on.** The reference selection above has near-dead efficiency at low momentum and at the large opening-angle extreme: from the fine truth distributions, the per-bin efficiency is only  $\sim 2\text{--}5\%$  for  $p_\mu, p_\pi < 0.15 \text{ GeV}/c$  (the 10/20 cm track-length cuts) and the region above the reco  $\theta_{\mu\pi} = 2.6 \text{ rad}$  cut is dominated by cosmics ( $\sim 7\%$  purity). The measured phase space  $p_\mu > 0.15, p_\pi > 0.175 \text{ GeV}/c, \theta_{\mu\pi} < 2.6 \text{ rad}$  (Section 6) raises and flattens the efficiency to  $\sim 13\text{--}20\%$  (Fig. 11). Table 7 compares the selection across the iterations. Two lessons: (i) the 2.6 rad *upper* cut is essential — dropping it admits  $3.4\times$  more cosmic background and cuts the purity to 45%; (ii) a *lower* opening-angle cut is counterproductive — the small-angle region is high-purity ( $\sim 58\%$ ), so cutting it lowers the selected signal without improving purity, which is why no lower cut is used.

Table 7: Selection performance across the phase-space iterations (NuWro fake data). “no upper cut” drops the  $\theta_{\mu\pi} < 2.6$  cut; “[0.314, 2.6]” adds a lower OA cut. The final choice keeps the 2.6 rad upper cut with no lower cut.

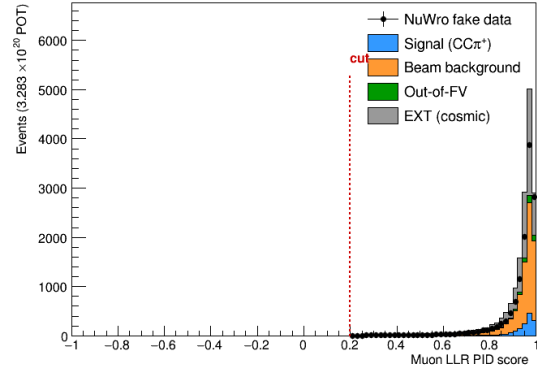
|                     | Reference<br>( $p > 0.1$ ) | no upper cut<br>( $p > .15/.175$ ) | [0.314, 2.6]<br>(lower cut) | <b>Final</b><br>( $\theta < 2.6$ ) |
|---------------------|----------------------------|------------------------------------|-----------------------------|------------------------------------|
| Generated signal    | 2684                       | 2055                               | 1916                        | 1999                               |
| Selected signal     | 324.7                      | 320.3                              | 301.8                       | 311.9                              |
| Efficiency          | 12.1%                      | 15.6%                              | 15.8%                       | <b>15.6%</b>                       |
| EXT (cosmic)        | 40.9                       | 139.0                              | 38.8                        | 40.9                               |
| Total bkg           | 243.6                      | 385.4                              | 252.3                       | 256.4                              |
| <b>Purity</b>       | <b>57.1%</b>               | 45.4%                              | 54.5%                       | <b>54.9%</b>                       |
| sig $\times$ purity | 185.6                      | 145.4                              | 164.4                       | <b>171.2</b>                       |

## 9.1 Key N−1 distributions

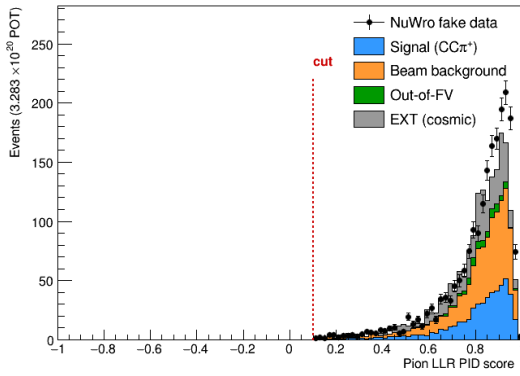
Figure 2 shows the N−1 (all cuts except the one under study) distributions for the four most discriminating variables. The arrows indicate the applied threshold.



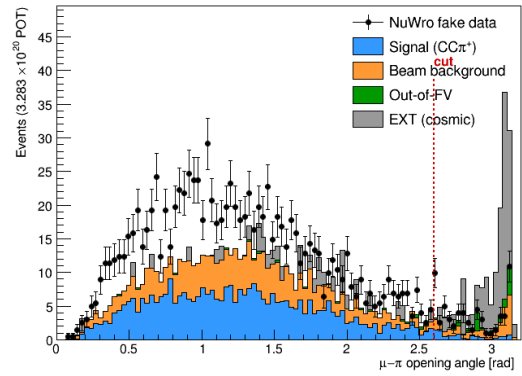
(a) Topological score (Cut 3).



(b) Muon LLR PID (Cut 4).



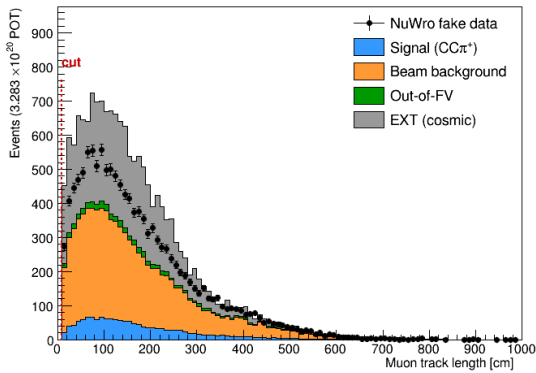
(c) Pion LLR PID (Cut 5).



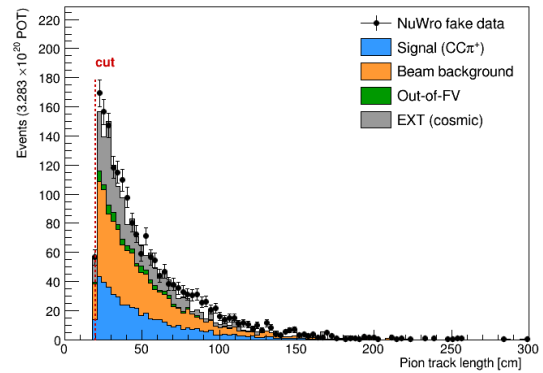
(d)  $\mu$ - $\pi$  opening angle (Cut 9).

Figure 2: N-1 distributions for four key selection cuts. Data (black points), stacked MC truth categories, and EXT (grey). Arrows indicate the applied cut threshold.

Figure 3 shows the N-1 distributions for the track-length cuts on both candidates.



(a) Muon candidate track length.

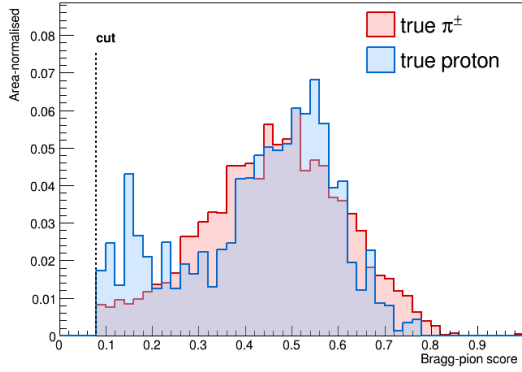


(b) Pion candidate track length.

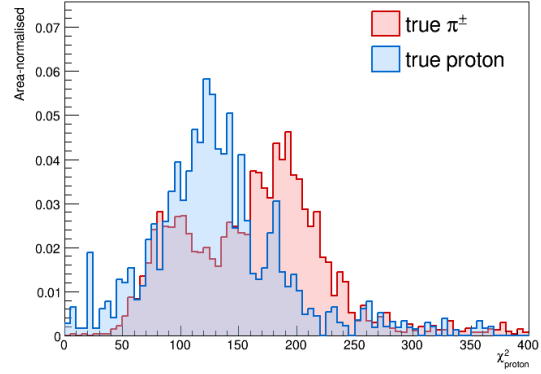
Figure 3: N-1 track-length distributions. Muon threshold:  $> 10$  cm (Cut 4); pion threshold:  $> 20$  cm (Cut 5).

## 9.2 Pion proton-rejection variables

Figure 4 shows the two main proton-rejection variables on the pion candidate, evaluated on MC truth-labelled pions ( $\pi^\pm$ ) and protons at the final cut stage, used to tune the working-point thresholds.



(a) Bragg-pion score (`trk_bragg_pion_v`). Working-point threshold:  $\geq 0.08$ .

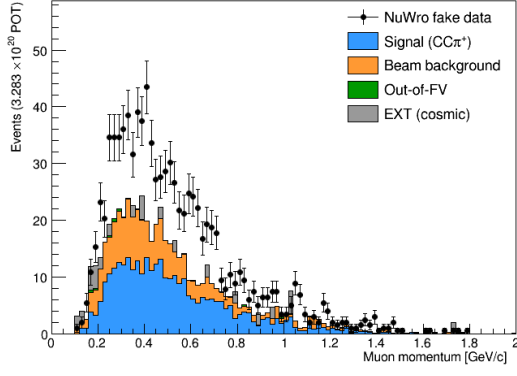


(b)  $\chi^2$  proton PID (`trk_pid_chipr_v`).

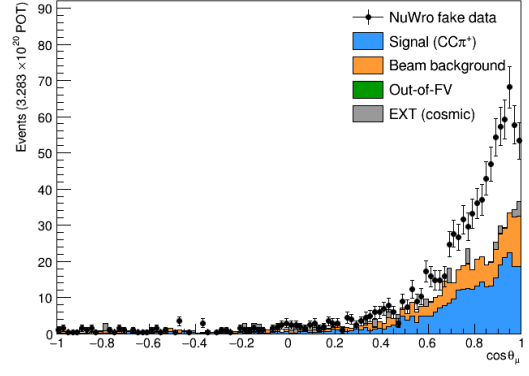
Figure 4: Pion-candidate PID distributions for true  $\pi^\pm$  (red) and true protons (blue) at the final cut stage. The Bragg-pion score provides the primary proton rejection;  $\chi_p^2$  is studied as cross-check.

## 9.3 Final-cut data/MC distributions

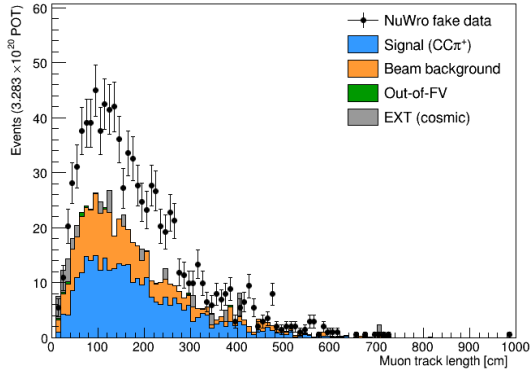
Figures 5 and 6 show the data/MC comparison for key variables at the final selection cut (Cut 10).



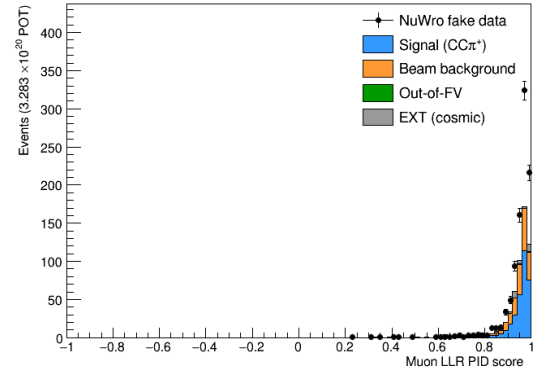
(a) Muon range/MCS momentum.



(b) Muon  $\cos \theta$ .

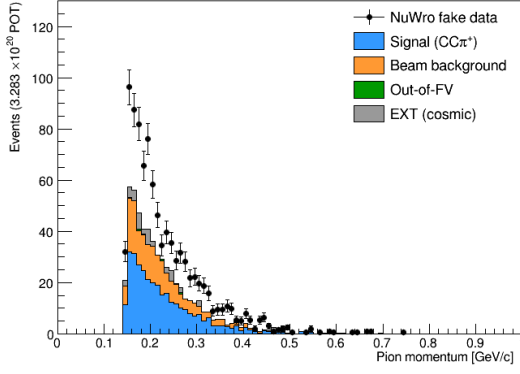


(c) Muon track length.

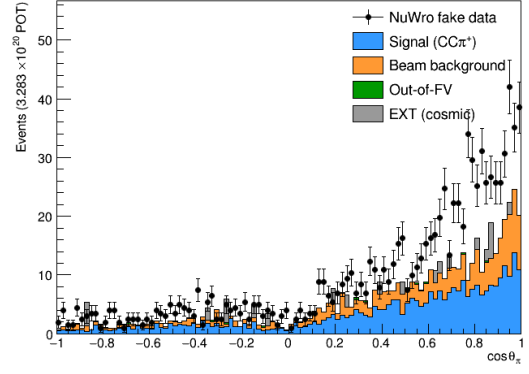


(d) Muon LLR PID.

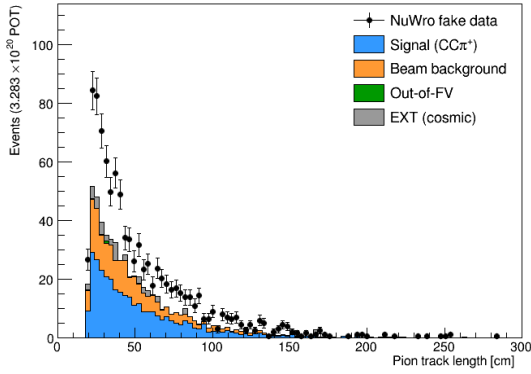
Figure 5: Data/MC comparison for muon-candidate variables at the final cut. Bottom panels show data/MC ratio. The overall data/MC = 0.75 deficit is visible; the shape in  $\cos \theta_\mu$  shows larger deficit at backward angles (residual cosmic contamination).



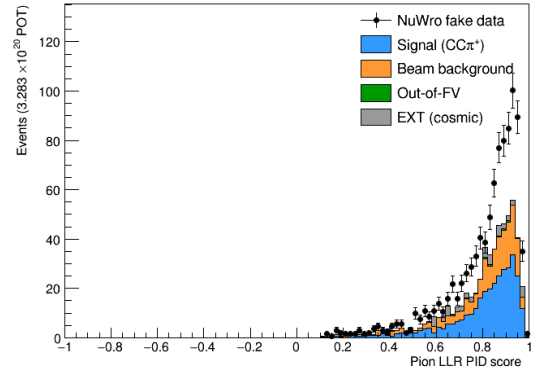
(a) Pion range momentum.



(b) Pion  $\cos \theta$ .

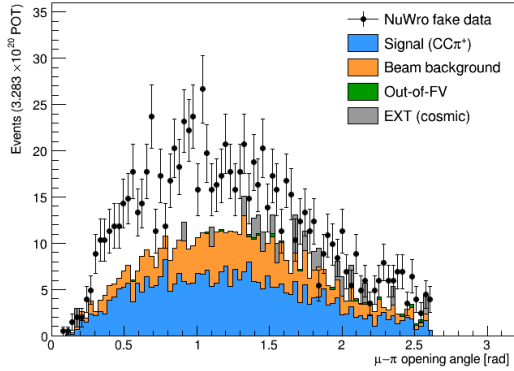


(c) Pion track length.

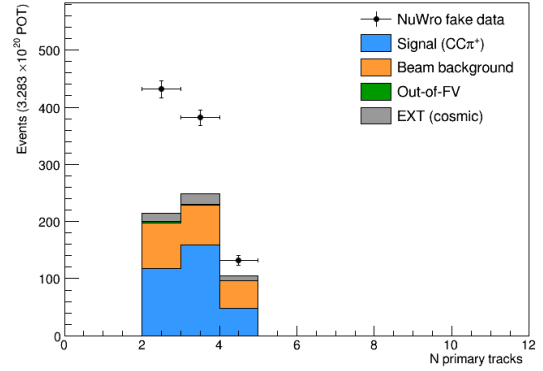


(d) Pion LLR PID.

Figure 6: Data/MC comparison for pion-candidate variables at the final cut.



(a)  $\mu$ - $\pi$  opening angle.



(b) Number of primary tracks.

Figure 7: Data/MC comparison for topological variables at the final cut.

## 9.4 Background decomposition

Table 8 shows the truth-level background breakdown at the final cut.

Table 8: Background composition at the final cut (truth-level MC categories, POT-weighted).

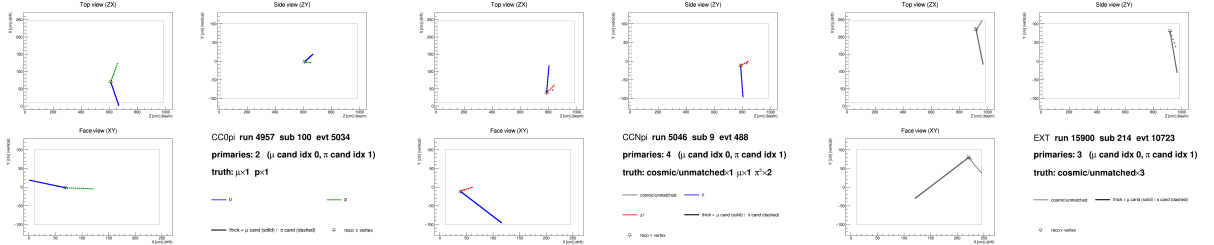
| Category                          | Events    | Fraction    |
|-----------------------------------|-----------|-------------|
| CC0 $\pi$ (proton mis-ID as pion) | $\sim 82$ | $\sim 41\%$ |
| CCN $\pi$ ( $\geq 2$ pion events) | $\sim 67$ | $\sim 34\%$ |
| OOFV (vertex outside FV)          | $\sim 18$ | $\sim 9\%$  |
| NC (neutral current)              | $\sim 14$ | $\sim 7\%$  |
| CC1 $\pi^0$                       | $\sim 8$  | $\sim 4\%$  |
| CC other / CC kaon                | $\sim 6$  | $\sim 3\%$  |
| $\nu_e$ / other flavour           | $\sim 4$  | $\sim 2\%$  |
| EXT (cosmic)                      | 40.9      | —           |
| Dirt                              | 3.8       | —           |

The dominant background ( $\sim 41\%$ ) is CC0 $\pi$  interactions in which a proton is reconstructed as the pion candidate. This mis-identification was verified directly by 2D event displays (truth-labelled by backtracked PDG): the pion candidate track shows a green (proton) truth colour with a stopping Bragg peak, while the genuine muon track is correctly identified.

At the current working point (`trk_bragg_pion_v`  $\geq 0.08$ ), the pion candidate is truth-matched to a proton in  $\approx 18\%$  of cases. Tighter Bragg-score thresholds (0.20–0.30) were studied but shown not to improve the figure of merit: the pion-to-proton ratio in CC $\pi^+$  signal is  $\approx 4 : 1$ , so rejecting protons removes genuine pions at nearly the same rate.

## 9.5 Event displays

Figure 8 shows representative 2D truth-labelled event displays for the two dominant backgrounds and the EXT cosmic sample. Tracks are coloured by backtracked truth PDG:  $\mu$  (blue),  $\pi^\pm$  (red),  $p$  (green), cosmic/unmatched (grey). The muon candidate is drawn solid-thick; the pion candidate dashed-thick; the reconstructed  $\nu$  vertex is shown as a star.



(a) CC0 $\pi$  background: a proton (green, dashed) is reconstructed as the pion candidate. (b) CCN $\pi$  background: two genuine charged pions present; one is selected as the pion candidate. (c) EXT cosmic event: all tracks are grey (no truth match), faking a neutrino vertex topology.

Figure 8: Truth-labelled 2D event displays (ZX projection) for representative background events at the final cut. Three views (ZX top, ZY side, XY face) are generated per event by `selection/event_display.C`.

## 10 Unfolding

### 10.1 Signal extraction

After the event selection, the background-subtracted reco-space signal spectrum for observable  $x$  is:

$$d_i^{\text{sig}} = d_i^{\text{data}} - b_i^{\nu\text{-bkg}} - b_i^{\text{EXT}} - b_i^{\text{dirt}}, \quad (11)$$



where  $i$  runs over reconstructed bins.

## 10.2 Response matrix

A 2D response matrix  $A_{ij}$  (true bin  $j$ , reco bin  $i$ ) is built from selected signal MC using the same binning on both axes. Each column is normalised to unit sum so that  $A_{ij} = P(\text{reco in bin } i | \text{true in bin } j)$ . The efficiency vector  $\varepsilon_j = \sum_i A_{ij}$  is the ratio of selected (`h_smear_sel`) to generated (`h_true_gen`) signal events per true bin.

## 10.3 Wiener-SVD unfolding

The primary method is Wiener-SVD [5], implemented in `xsec/WienerSVD.h` (`RunWienerSVD_FW`) as a faithful replica of the official `XSecAnalyzer WienerSVDUnfolder`: Cholesky whitening of the data covariance, regularisation inside the SVD of  $RC^{-1}$ , and the BNL filter (Eq. 3.24 of Ref. [5]). The method applies a Wiener filter in the SVD basis, providing adaptive regularisation whose strength is set by the data covariance. Because the custom pipeline does not throw systematic universes, the covariance is imported from the framework (the prediction covariance scaled to the custom prediction, plus the custom data Poisson; Section 12); a stat-only covariance under-regularises and over-corrects the integral by  $\sim 26\%$ .

The differential cross section is:

$$\left. \frac{d\sigma}{dx} \right|_i = \frac{\hat{N}_i^{\text{unfold}}}{\varepsilon_i \cdot \Delta x_i \cdot \Phi_{\text{total}} \cdot N_{\text{Ar}}}, \quad (12)$$

where  $\hat{N}_i$  is the Wiener-SVD unfolded signal count in true bin  $i$ ,  $\Phi_{\text{total}} = \Phi_{\text{per POT}} \times T_{\text{POT}}$  is the total integrated flux, and  $N_{\text{Ar}}$  is the number of argon target nuclei in the FV. Equivalently, the events-to-cross-section conversion is  $\text{conv} = N_{\text{Ar}} \Phi / 10^{-38}$  (so  $\sigma_i = \hat{N}_i / (\varepsilon_i \text{conv})$ ), with the authoritative flux constant  $\phi = 6.81159 \times 10^{-10} \text{ cm}^{-2}/\text{POT}$  ( $E > 60 \text{ MeV}$ ; Sec. 4). At the fake-data POT the inputs are:

| Quantity                                        | Value                                                |
|-------------------------------------------------|------------------------------------------------------|
| $N_{\text{Ar}}$ (FV, dead-region excluded)      | $8.710 \times 10^{29}$                               |
| Flux constant $\phi$ ( $E > 60 \text{ MeV}$ )   | $6.81159 \times 10^{-10} \text{ cm}^{-2}/\text{POT}$ |
| POT                                             | $7.630913 \times 10^{20}$                            |
| Integrated flux $\Phi = \phi \times \text{POT}$ | $5.198 \times 10^{11} \text{ cm}^{-2}$               |
| $\text{conv} = N_{\text{Ar}} \Phi / 10^{-38}$   | $4.527 \times 10^3$                                  |

An integrated cross section is computed as the width-weighted sum,

$$\sigma_{\text{int}} = \sum_i \left. \frac{d\sigma}{dx} \right|_i \Delta x_i, \quad (13)$$

and serves as a closure check: all five observables must agree.

## 10.4 Reconstructed spectra and response matrices

Figure 9 shows the background-subtracted reco-space signal spectra for all five observables, overlaid with the MC signal prediction and the individual background components.

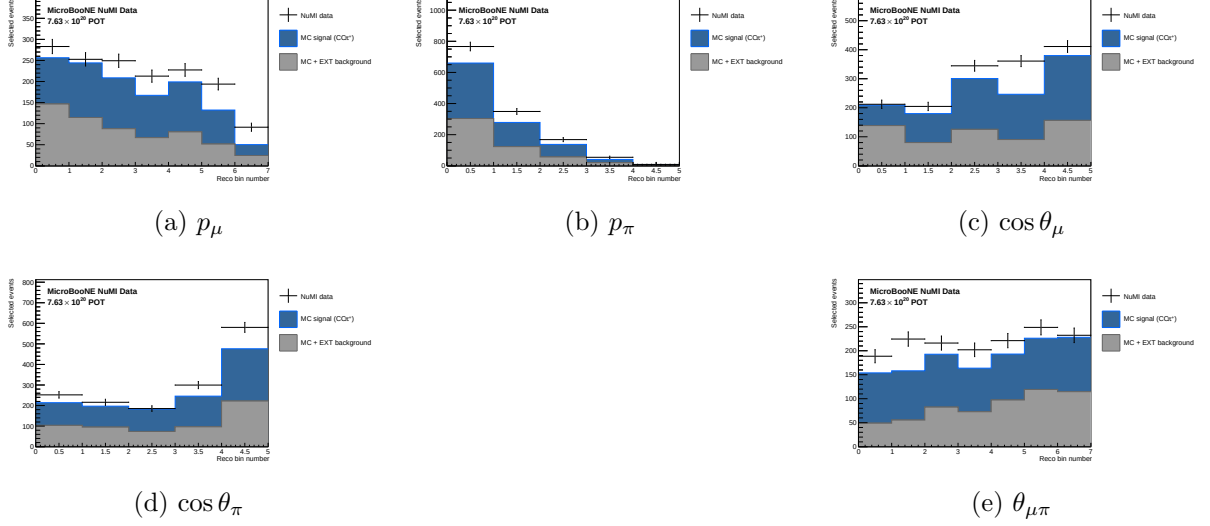


Figure 9: Reconstructed-space spectra at the final cut for all five observables. NuWro fake data (black points), stacked signal (blue), beam background (orange), dirt (green), and EXT (grey). NuWro carries roughly twice the GENIE-predicted signal, consistent with the generator spread in  $CC\pi^+$  production.

Figure 10 shows the response matrices (normalised per true bin) for the five observables. The diagonal dominance confirms that migration effects are modest for the angular observables;  $p_\mu$  shows slightly larger migrations in the low-momentum bins.

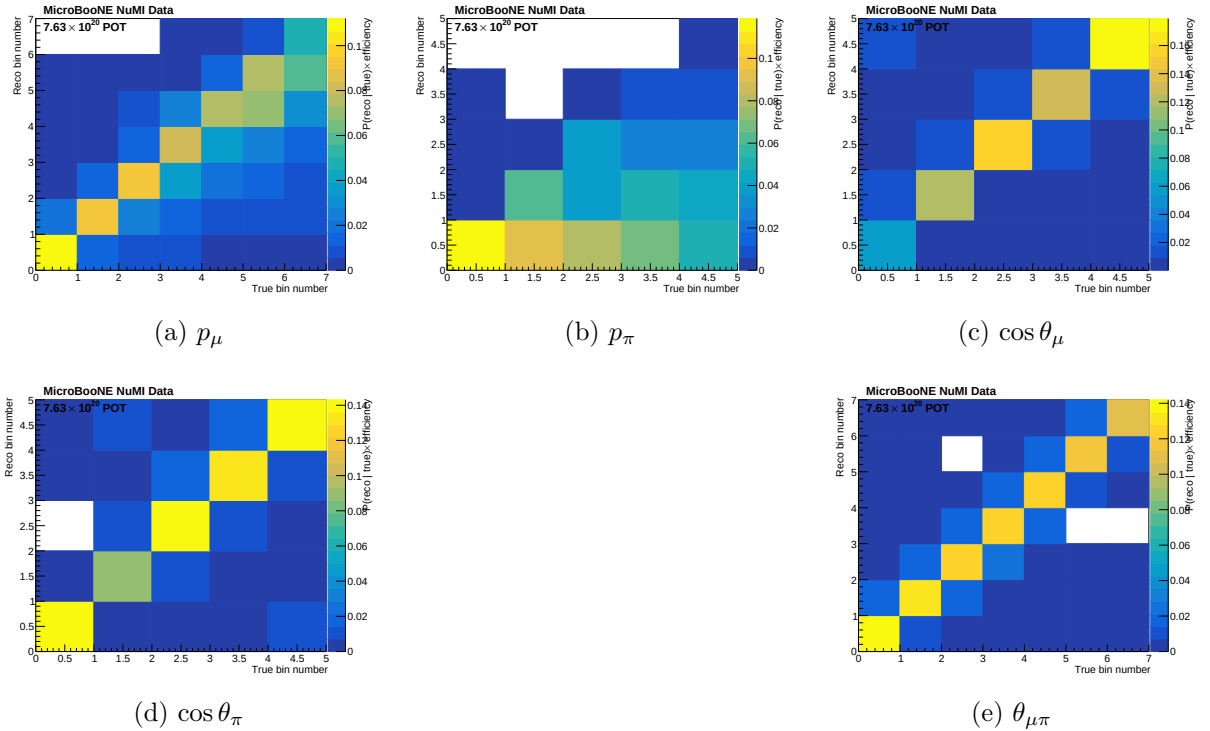


Figure 10: Response matrices  $A_{ij}$  for the five observables. Each column is normalised to unity (migration probability). Colour scale: white = 0, dark blue = 1.

Figure 11 shows the selection efficiency per true bin for all five observables, computed as the ratio of selected to generated signal MC.

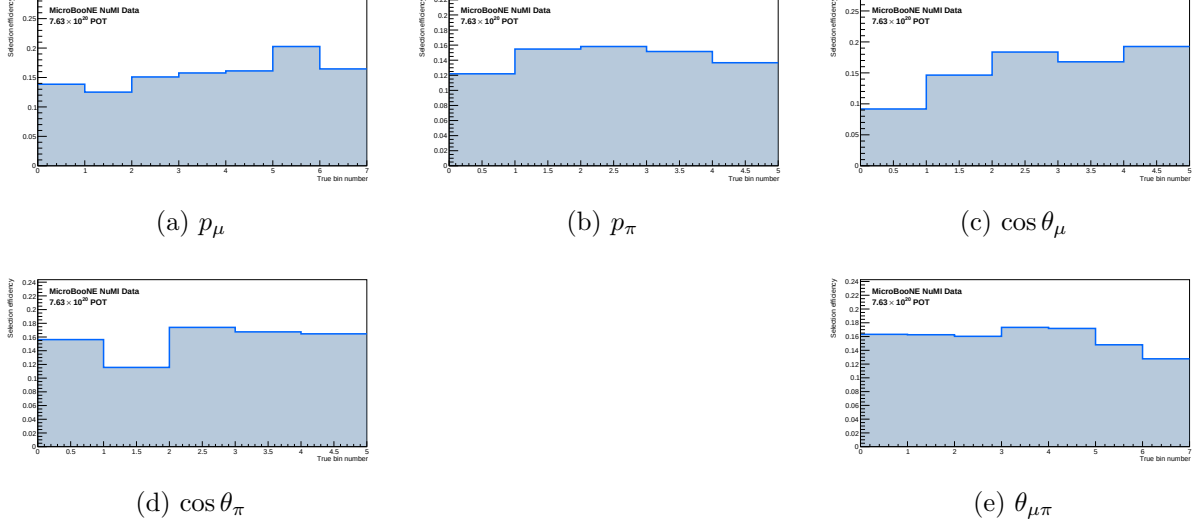


Figure 11: Selection efficiency as a function of true observable value, in the measured phase space ( $p_\mu > 0.15$ ,  $p_\pi > 0.175$  GeV/c,  $\theta_{\mu\pi} < 2.6$  rad). Efficiency is the ratio of selected to generated signal MC ( $\varepsilon_i = \sum_j A_{ij}$ , integrated over reco bins). With the momentum thresholds and the 2.6 rad opening-angle cut the efficiency is flat at  $\sim 13$ – $20\%$  across all observables; the former  $\sim 5\%$  low-momentum turn-on bins and the cosmic-dominated large-angle region are removed. The lowest point is the smallest opening-angle bin ( $\sim 9\%$  efficiency) — it is retained because it is high-purity ( $\sim 58\%$ ), not low-purity.

## 10.5 Reconstruction resolution

Figure 12 shows the reconstruction resolution of each measured observable, evaluated on selected signal events (GENIE  $\nu_\mu$  overlay, measured phase space). For the momenta the relative resolution  $(\text{reco} - \text{true})/\text{true}$  is shown; for the angular observables the absolute residual  $(\text{reco} - \text{true})$  is used, since the relative metric diverges as  $\cos \theta \rightarrow 0$  /  $\theta_{\mu\pi} \rightarrow 0$ . The red points are the mean residual per true bin; the quoted numbers are the overall mean (bias) and RMS (resolution).

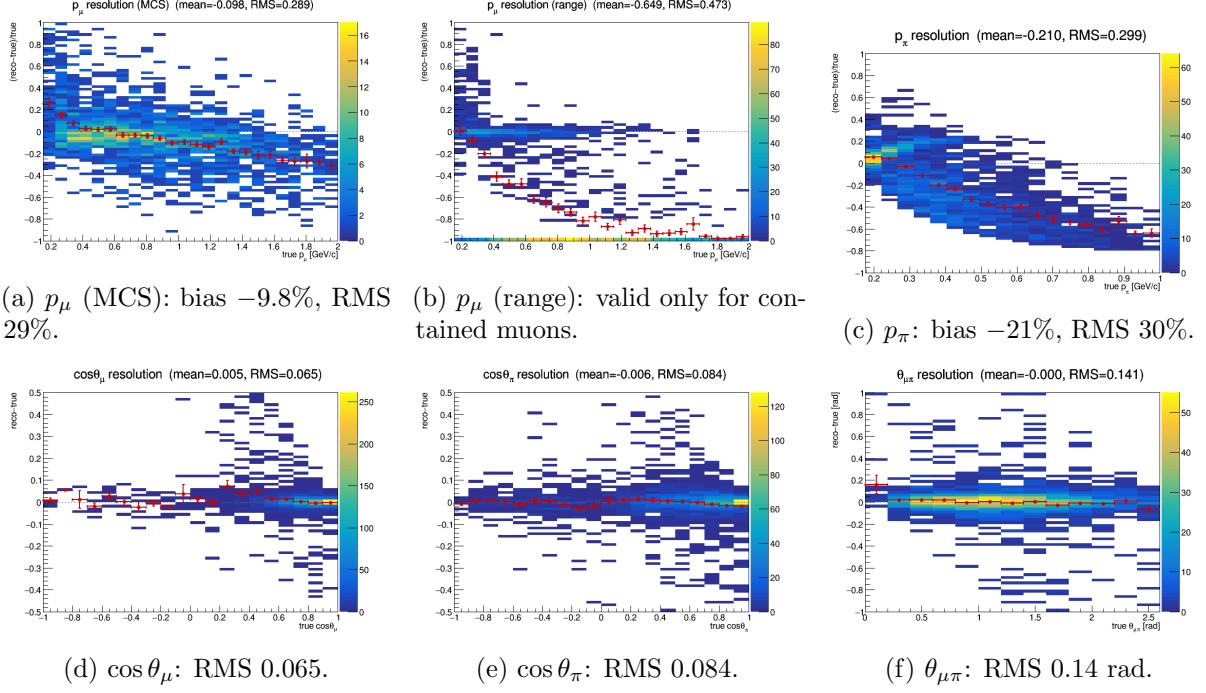


Figure 12: Reconstruction resolution vs. true value, selected signal. Momenta: relative ( $\text{reco} - \text{true}$ )/ $\text{true}$ ; angles: absolute ( $\text{reco} - \text{true}$ ). Red points: mean residual per true bin. The muon MCS momentum slightly over-estimates at low  $p_\mu$  and increasingly under-estimates above  $\sim 1$  GeV/ $c$ ; the range estimator is badly biased for *exiting* muons (range = distance to wall), which is why the analysis uses range only for contained muons (range-or-MCS). The pion momentum (proton-hypothesis range KE) under-estimates true  $p_\pi$  by  $\sim 21\%$ . The angular variables are essentially unbiased with narrow resolution.

## 10.6 Cross-check methods

Seven additional methods are run in parallel:

Table 9: Unfolding methods.

| Label                     | Description                                        |
|---------------------------|----------------------------------------------------|
| <code>wiener_svd</code>   | Wiener-SVD (primary result)                        |
| <code>bin_by_bin</code>   | Efficiency correction only; no migration unfolding |
| <code>ibu</code>          | D’Agostini iterative Bayesian, 4 iterations        |
| <code>tikhonov</code>     | Tikhonov-regularised least-squares, $\tau = 1.0$   |
| <code>roo_bayes</code>    | RooUnfoldBayes, 4 iterations                       |
| <code>roo_svd</code>      | RooUnfoldSvd                                       |
| <code>roo_invert</code>   | RooUnfoldInvert (direct matrix inversion)          |
| <code>roo_binbybin</code> | RooUnfoldBinByBin                                  |

## 11 Results

### 11.1 Integrated cross section

The cross sections below are reported in the *measured phase space*  $p_\mu > 0.15$  GeV/ $c$ ,  $p_\pi > 0.175$  GeV/ $c$ ,  $\theta_{\mu\pi} < 2.6$  rad (Section 6), extracted with the framework-consistent Wiener-SVD

unfolder using the imported systematic covariance (Section 12). Figure 13 and Table 10 give the integrated cross section for each observable on the NuWro fake-data sample.

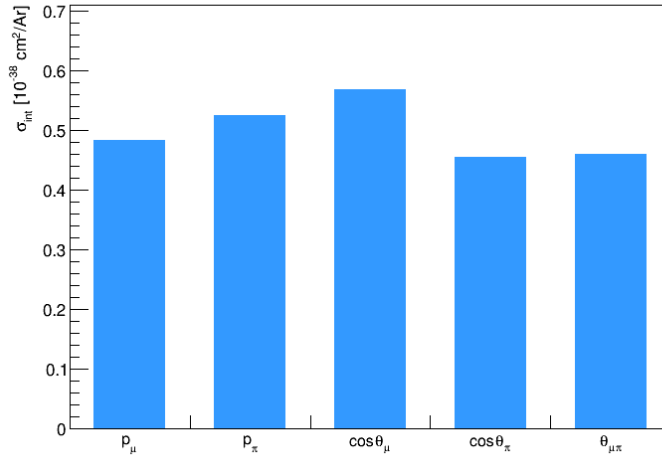


Figure 13: Integrated cross section  $\sigma_{\text{int}}$  from each of the five observables (framework Wiener-SVD, optimised binning, GENIE detVar-CV fake data). Total uncertainty (full covariance); optimised binning. The values differ between observables because each integrates a different projection of the same restricted phase space.

Table 10: Integrated cross section from each observable (framework Wiener-SVD, optimised binning, measured phase space  $p_\mu > 0.15$ ,  $p_\pi > 0.175 \text{ GeV}/c$ ,  $\theta_{\mu\pi} < 2.6 \text{ rad}$ ), GENIE detVar-CV fake data. Units  $10^{-38} \text{ cm}^2/\text{Ar}$ ; total uncertainty (full covariance). Flux:  $\Phi = 6.81159 \times 10^{-10} \text{ cm}^{-2} \text{ POT}^{-1}$  (Eq. 4).

| Observable                  | $\sigma_{\text{int}}$ |
|-----------------------------|-----------------------|
| $d\sigma/dp_\mu$            | 1.137                 |
| $d\sigma/dp_\pi$            | 1.125                 |
| $d\sigma/d \cos \theta_\mu$ | 1.159                 |
| $d\sigma/d \cos \theta_\pi$ | 0.931                 |
| $d\sigma/d\theta_{\mu\pi}$  | 1.262                 |

**Relation to the framework cross-check.** Before the signal-definition refinement, the custom and official XSecAnalyzer pipelines were validated against each other at a common *reference* phase space ( $p_\mu, p_\pi > 0.1 \text{ GeV}/c$ , full opening angle with the reco  $\theta_{\mu\pi} < 2.6 \text{ rad}$  cut in place): with the imported systematic covariance, the EXT cancellation for fake data, and matched binning, the two fully independent pipelines agreed to  $\leq 2.1\%$  on every observable ( $p_\mu$  1.016,  $p_\pi$  0.979,  $\cos \theta_\mu$  0.987,  $\cos \theta_\pi$  1.004,  $\theta_{\mu\pi}$  1.001). That established the correctness of the selection, response, background subtraction, and unfolding. The results reported below are the XSECANALYZER framework extraction at the *measured phase space* ( $p_\mu > 0.15$ ,  $p_\pi > 0.175 \text{ GeV}/c$ ,  $\theta_{\mu\pi} < 2.6 \text{ rad}$ , chosen from the efficiency turn-on curves, Sec. 17) on the resolution-optimised binning; the efficiency flattens from 12.1% to 15.6% at  $\sim 55\%$  purity (the 2.6 rad cut rejects the large-angle cosmic background; no lower opening-angle cut is used, as the small-angle region is high-purity). The fully independent custom pipeline reproduces the framework and now serves as the cross-check (Sec. 16.1).

## 11.2 Differential cross section: $d\sigma/dp_\mu$

Table 11: Framework Wiener-SVD  $d\sigma/dp_\mu$  [ $10^{-38}$  cm<sup>2</sup>/(GeV/c)/Ar] (full framework covariance). Total uncertainty (full covariance); optimised binning. Flux: Eq. 4.

| Bin | $p_\mu$ range<br>(GeV/c) | $d\sigma/dp_\mu$ | Total unc.  |
|-----|--------------------------|------------------|-------------|
| 1   | [0.15, 0.35]             | +0.826           | $\pm 0.229$ |
| 2   | [0.35, 0.55]             | +0.820           | $\pm 0.221$ |
| 3   | [0.55, 0.75]             | +0.818           | $\pm 0.209$ |
| 4   | [0.75, 0.95]             | +0.812           | $\pm 0.197$ |
| 5   | [0.95, 1.25]             | +0.537           | $\pm 0.127$ |
| 6   | [1.25, 1.75]             | +0.321           | $\pm 0.075$ |
| 7   | [1.75, 3.00]             | +0.128           | $\pm 0.030$ |

The spectrum falls smoothly above  $\sim 0.5$  GeV/c. The first bin sits high because it is narrow (0.05 GeV/c) and near the acceptance edge. The broad bins at  $p_\mu > 2$  GeV/c reduce statistical fluctuations in the sparse high-momentum tail.

Figure 14 shows the custom  $d\sigma/dp_\mu$  and the background-subtraction diagnostic.

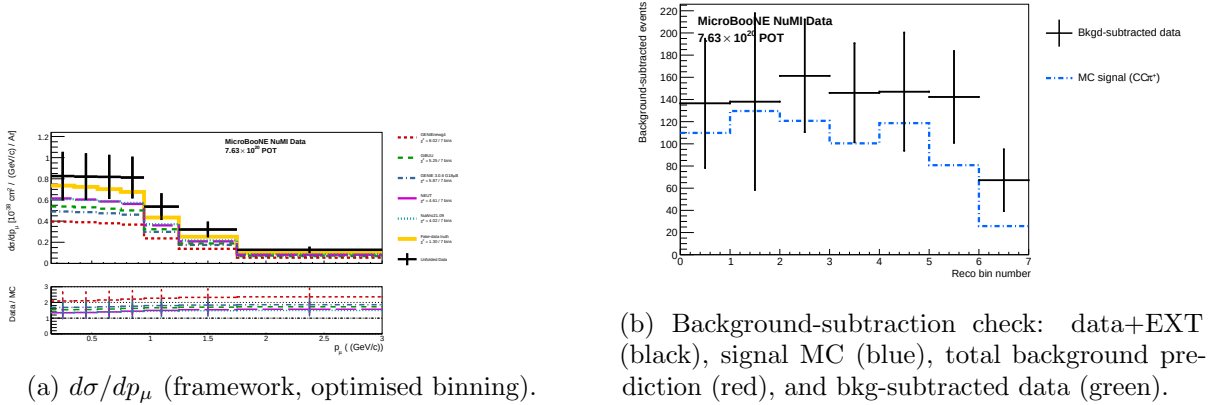


Figure 14: Muon momentum  $d\sigma/dp_\mu$  and the background-subtraction diagnostic.

## 11.3 Differential cross section: $d\sigma/dp_\pi$

Table 12: Framework Wiener-SVD  $d\sigma/dp_\pi$  [ $10^{-38}$  cm<sup>2</sup>/(GeV/c)/Ar]. Total uncertainty (full covariance); optimised binning. Flux: Eq. 4.

| Bin | $p_\pi$ range<br>(GeV/c) | $d\sigma/dp_\pi$ | Total unc.  |
|-----|--------------------------|------------------|-------------|
| 1   | [0.17, 0.25]             | +2.998           | $\pm 0.811$ |
| 2   | [0.25, 0.32]             | +3.201           | $\pm 0.874$ |
| 3   | [0.32, 0.42]             | +2.249           | $\pm 0.604$ |
| 4   | [0.42, 0.55]             | +1.736           | $\pm 0.465$ |
| 5   | [0.55, 1.00]             | +0.502           | $\pm 0.135$ |

The spectrum is broad across  $p_\pi \lesssim 0.5$  GeV/c, consistent with pions from  $\Delta(1232)$  decays. The first bin is a narrow 0.025 GeV/c sliver at the acceptance threshold, hence its large  $d\sigma/dp_\pi$  and

uncertainty.

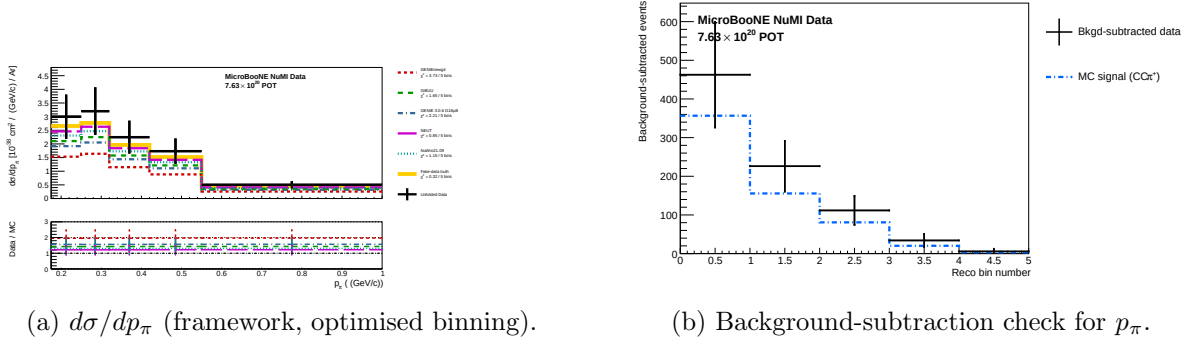


Figure 15: Pion momentum  $d\sigma/dp_\pi$  unfolding diagnostics.

#### 11.4 Differential cross section: $d\sigma/d\cos\theta_\mu$

Table 13: Framework Wiener-SVD  $d\sigma/d\cos\theta_\mu$  [ $10^{-38}$  cm<sup>2</sup>/Ar]. With the EXT (cosmic) subtraction cancelled for the NuWro fake data (Section 12), all bins are now positive, including the backward-muon region. Total uncertainty (full covariance); optimised binning.

| Bin | $\cos\theta_\mu$ range | $d\sigma/d\cos\theta_\mu$ | Total unc.  |
|-----|------------------------|---------------------------|-------------|
| 1   | $[-1.00, 0.45]$        | +0.143                    | $\pm 0.045$ |
| 2   | $[0.45, 0.65]$         | +1.082                    | $\pm 0.298$ |
| 3   | $[0.65, 0.80]$         | +1.500                    | $\pm 0.400$ |
| 4   | $[0.80, 0.90]$         | +2.546                    | $\pm 0.601$ |
| 5   | $[0.90, 1.00]$         | +2.561                    | $\pm 0.620$ |

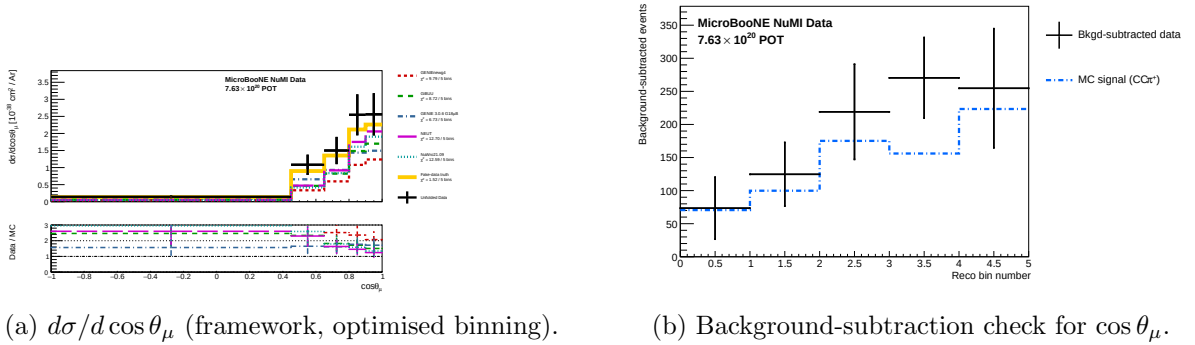
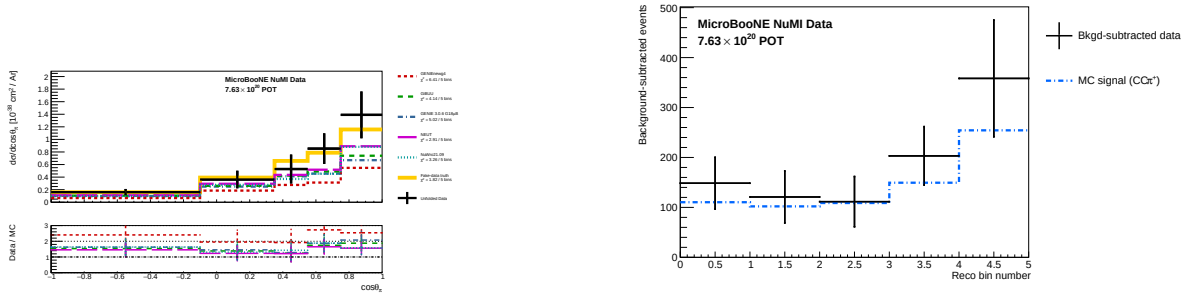


Figure 16: Muon polar angle  $d\sigma/d\cos\theta_\mu$  (framework, optimised binning) and the background-subtraction diagnostic. The spectrum rises steeply toward forward angles.

## 11.5 Differential cross section: $d\sigma/d\cos\theta_\pi$

Table 14: Framework Wiener-SVD  $d\sigma/d\cos\theta_\pi$  [ $10^{-38}$  cm<sup>2</sup>/Ar]. Total uncertainty (full covariance); optimised binning. Flux: Eq. 4.

| Bin | $\cos\theta_\pi$ range | $d\sigma/d\cos\theta_\pi$ | Total unc.  |
|-----|------------------------|---------------------------|-------------|
| 1   | $[-1.00, -0.10]$       | +0.161                    | $\pm 0.051$ |
| 2   | $[-0.10, 0.35]$        | +0.362                    | $\pm 0.140$ |
| 3   | $[0.35, 0.55]$         | +0.529                    | $\pm 0.231$ |
| 4   | $[0.55, 0.75]$         | +0.853                    | $\pm 0.244$ |
| 5   | $[0.75, 1.00]$         | +1.390                    | $\pm 0.372$ |



(a)  $d\sigma/d\cos\theta_\pi$  (framework, optimised binning).

(b) Background-subtraction check for  $\cos\theta_\pi$ .

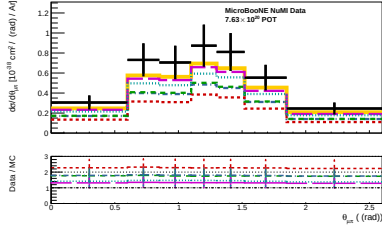
Figure 17: Pion polar angle  $d\sigma/d\cos\theta_\pi$  (framework, optimised binning) and the background-subtraction diagnostic.

## 11.6 Differential cross section: $d\sigma/d\theta_{\mu\pi}$

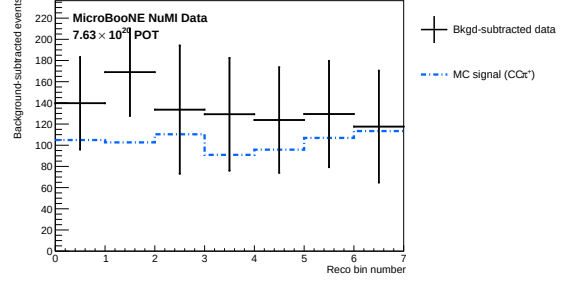
Table 15: Framework Wiener-SVD  $d\sigma/d\theta_{\mu\pi}$  [ $10^{-38}$  cm<sup>2</sup>/rad/Ar], in the measured range  $\theta_{\mu\pi} < 2.6$  rad (9 bins; no lower cut — the small-angle region is retained as it is high-purity, Section 6). The spectrum peaks at  $\theta_{\mu\pi} \approx 0.9$ –1.3 rad. The narrow last bin  $[2.513, 2.6]$  has a correspondingly larger  $d\sigma/d\theta_{\mu\pi}$  and uncertainty. Total uncertainty (full covariance); optimised binning.

| Bin | $\theta_{\mu\pi}$ range (rad) | $d\sigma/d\theta_{\mu\pi}$ | Total unc.  |
|-----|-------------------------------|----------------------------|-------------|
| 1   | $[0.00, 0.60]$                | +0.305                     | $\pm 0.070$ |
| 2   | $[0.60, 0.85]$                | +0.731                     | $\pm 0.165$ |
| 3   | $[0.85, 1.10]$                | +0.705                     | $\pm 0.167$ |
| 4   | $[1.10, 1.30]$                | +0.873                     | $\pm 0.210$ |
| 5   | $[1.30, 1.52]$                | +0.811                     | $\pm 0.189$ |
| 6   | $[1.52, 1.85]$                | +0.553                     | $\pm 0.129$ |
| 7   | $[1.85, 2.60]$                | +0.245                     | $\pm 0.059$ |





(a)  $d\sigma/d\theta_{\mu\pi}$  (framework, optimised binning).



(b) Background-subtraction check for  $\theta_{\mu\pi}$ .

Figure 18: Opening angle  $d\sigma/d\theta_{\mu\pi}$  unfolding diagnostics.

## 12 Data/MC Normalisation

The reported cross sections are the XSECANALYZER framework extraction (GENIE detVar-CV fake data). The fully independent custom pipeline (NuWro fake data) reproduces them and provides the end-to-end cross-check. Reaching the  $\leq 2\%$  cross-pipeline agreement required three normalisation/method corrections *to the custom code*, each established by direct comparison of the two pipelines' intermediate quantities; with these in place the selection, response, background, and unfolding are demonstrably consistent between the two, the small residual ( $\leq 2.1\%$ ) dominated by the covariance-import approximation (below).

### 12.1 Unfolding covariance ( $\sim 26\%$ effect)

Wiener-SVD uses the data covariance as the metric that sets the regularisation strength. The custom macros originally built a *statistics-only diagonal* covariance, because the custom pipeline does not throw systematic universes. With the  $\text{CC}\pi^+$  response strongly off-diagonal (50–99% bin migration), a stat-only covariance under-states the noise, under-regularises, and inflates the unfolded integral by  $\sim 26\%$ . This was proven by a swap test: feeding the framework's own inputs through the same Wiener-SVD unfolders with a stat-only covariance reproduced the inflated result, while its full systematic covariance recovered the NuWro truth.

The fix imports the framework's covariance into the custom unfolders. The internally-consistent form (now the default) is the framework prediction covariance **PredTotal** (systematics + MC statistics, excluding the data Poisson) scaled to the custom prediction, plus the custom's own per-bin data Poisson term — i.e. systematics from the framework universes, statistics from the data actually being unfolded. This removes the 26% bias and brings all five observables to  $\leq 2\%$  of the framework.

### 12.2 EXT (cosmic) subtraction for fake data

The NuWro fake-data sample is pure Monte Carlo and contains *no* beam-off (cosmic) contamination, so the EXT sample must not be net-subtracted from it. The framework handles this by *adding* the EXT histogram to the fake-data spectrum, so the EXT term cancels in  $\text{data}_{\text{signal}} = (\text{data} + \text{EXT}) - (\text{MC bkg} + \text{EXT} + \text{dirt})$ . The custom macros originally subtracted EXT ( $\approx 41$  events at  $3.283 \times 10^{20}$  POT) from a sample that never contained it, a flat  $\sim 3.7\%$  deficit on every observable. Replicating the framework's EXT cancellation (`fake_data_ext_cancel` in `ccpi_xsec.C`) removes it. For *real* beam-on data this term must be re-enabled, since real data does contain cosmics.

### 12.3 Opening-angle binning (review item C4)

The custom  $\theta_{\mu\pi}$  binning used 10 reco/true bins; the framework uses 9 (the last two custom bins merged into  $[2.513, 3.142]$ ). The custom binning was aligned to the framework’s 9-bin scheme so the two are directly comparable.

### 12.4 Generator normalisation (NuWro vs GENIE)

In the closure test the NuWro fake data carries roughly twice the selected  $\text{CC}\pi^+$  signal of the GENIE central-value prediction used to build the response, consistent with the well-documented generator spread in  $\text{CC}\pi^+$  production on argon [9]. The unfolding correctly recovers the injected NuWro truth (Section 16), confirming that the  $\sim 2\times$  generator difference is propagated faithfully rather than absorbed into an efficiency or normalisation artefact.

### 12.5 Software trigger (verified to be a no-op)

A hypothesis that the absence of a `swtrig == 1` filter on MC events was causing a  $\sim 13\%$  over-normalisation was investigated. The cut was implemented and the full pipeline rerun; the result was byte-identical. All MC events passing the topological cut already carry `swtrig == 1`: the  $\sim 13\%$  of raw MC events failing `swtrig` also fail the topological cut. The cut has been retained for formal correctness but contributes nothing to the current normalisation. Likewise, adding the GENIE spline weight to the central-value weight was a verified no-op: `weightSpline`  $\equiv 1$  in the NuMI ntuples and the NuWro overlay’s CV weights are sentinel ( $-1$ ) values clamped to unity.

## 13 Systematic Uncertainties

Full systematic uncertainty propagation has not yet been performed. Table 16 lists the sources and their estimated impacts.

Table 16: Systematic uncertainty sources. “Not quantified” means the branch is readable but propagation code has not been written.

| Source                        | Branch / method           | Impact                              |
|-------------------------------|---------------------------|-------------------------------------|
| Flux (PPFX multisims)         | <code>weightsFlux</code>  | $\sim 10\text{--}15\%$ norm.        |
| GENIE cross sections          | <code>weightsGenie</code> | $\sim 20\text{--}30\%$ (model)      |
| Hadron re-interaction         | <code>weightsReint</code> | $\sim 5\text{--}10\%$               |
| Detector response             | Dedicated samples         | $\sim 5\text{--}15\%$               |
| EXT normalisation             | Trigger-count ratio       | $\sim 3\text{--}5\%$                |
| POT counting                  | Beam toroids              | $\sim 2\%$                          |
| Flux integral (absolute)      | PPFX integral             | Blocks absolute $\sigma$            |
| Statistical (427 data events) | —                         | $\sim 5\%$ on $\sigma_{\text{int}}$ |

## 14 Known Issues and Required Follow-up

Table 17: Outstanding issues and their impact on the final result.

| Issue                                                              | Status                                  | Impact                                                                                         |
|--------------------------------------------------------------------|-----------------------------------------|------------------------------------------------------------------------------------------------|
| FLUX_PER_POT confirmed                                             | <b>Resolved</b>                         | $\Phi = 6.81159 \times 10^{-10} \text{ cm}^{-2} \text{ POT}^{-1}$<br>from NuMI flux histograms |
| Unfolding covariance (stat-only $\rightarrow$ full)                | <b>Resolved</b>                         | Framework <code>PredTotal</code> +data<br>Poisson imported; removed<br>$\sim 26\%$ bias        |
| EXT subtraction on NuWro fake data                                 | <b>Resolved</b>                         | Cancelled via <code>fake_data_ext_cancel</code> ;<br>re-enable for real data                   |
| $\theta_{\mu\pi}$ binning (10 vs 9, C4)                            | <b>Resolved</b>                         | Custom aligned to framework<br>9-bin scheme                                                    |
| Multisim systematics not propagated (custom)                       | Not started                             | Imported from framework for<br>the covariance; full custom<br>propagation pending              |
| Negative tail bins                                                 | <b>Resolved</b>                         | All bins now positive after EXT<br>cancellation; no longer present                             |
| Runs 2, 4, 5 beam-on files not available                           | Data files needed                       | $\times 5.4$ exposure increase for the<br>real-data measurement                                |
| Generator comparisons (GENIE, NuWro)                               | Blocked by missing AVX on compute nodes | NEUT comparison runnable<br>locally                                                            |
| CRT veto branches ( <code>crthitpe</code> , <code>crtveto</code> ) | Dead (all-zero in NuMI ntuples)         | Replaced by <code>bdt_cosmic</code><br>(disabled at default 0.0)                               |
| Official XSecAnalyzer NuWro closure                                | <b>Done</b> (all 5 obs)                 | Cross-validates the unfolding;<br>NuWro $\approx 2\times$ GENIE (Sec. 16)                      |

## 15 Analysis Pipeline

The primary measurement is produced by a single driver, `run_analysis.sh`, which chains the selection, the per-observable cross-section extraction, and the comparison/diagnostic stages:

```

selection/ccpi_selection.C  ->  histograms.root + unfolding_inputs.root
                             | (5 observables x 8 methods)
xsec/ccpi_xsec.C(obs,method) -> xsec_{obs}_{method}.root  (40 files)
xsec/compare_unfolding.C    -> compare_plots/
xsec/compare_generators.C   -> gen_compare/
selection/event_display.C    -> event_displays/

```

Each output file is validated for freshness and non-zero content after the step that produces it. The official-framework cross-check of Section 16 is driven separately (`xsec_analyzer/run_nuwro_observables.sh`).

## 16 NuWro Fake-Data Closure and Cross-Pipeline Validation

As a generator-model closure and an independent cross-check of the extraction, the XSECANALYZER framework used throughout this note was additionally run (`ProcessNTuples`  $\rightarrow$  `UniverseMaker`  $\rightarrow$  `Unfold`) with a NuWro overlay as *fake data* in place of the GENIE detVar-CV sample of the main results. The NuWro reconstructed spectrum is unfolded with the GENIE detector response, efficiency, and GENIE-predicted background subtraction, then compared to the GENIE MicroBooNE-tune

prediction. If the unfolding is unbiased, the ratio reflects the genuine NuWro-versus-GENIE model difference rather than a procedural artefact.

Extending the framework beyond  $p_\mu$  required six additional output branches in the `CC1mu1piXp` selection (reconstructed and true pion momentum, muon  $\cos\theta$ , and pion  $\cos\theta$ ); the reconstructed pion momentum uses the proton-hypothesis range kinetic energy (`trk_energy_proton_v`), matching the custom pipeline. The true  $\theta_{\mu\pi}$  opening angle, previously written as all-zeros, was also corrected, and all fourteen samples were re-processed.

Table 18: Configuration of the NuWro fake-data closure.

| Item          | Value                                                                                    |
|---------------|------------------------------------------------------------------------------------------|
| Fake data     | NuMI FHC Run 1 NuWro overlay, $6.6504 \times 10^{20}$ POT (onBNB)                        |
| Exposure      | all MC/EXT/dirt/detVar scaled to NuWro POT (high-statistics closure)                     |
| Response/eff. | GENIE $\nu_\mu$ MC overlay (central value)                                               |
| Unfolding     | Wiener-SVD, second-derivative regularisation                                             |
| Systematics   | full multisim + detector-variation universes                                             |
| Observables   | all five ( $p_\mu$ , $p_\pi$ , $\cos\theta_\mu$ , $\cos\theta_\pi$ , $\theta_{\mu\pi}$ ) |

The NuWro fake data unfolds to approximately twice the GENIE MicroBooNE tune in every observable, uniformly in shape (Fig. 19, Table 19). This *reproduces* the primary-pipeline finding (Section 12), cross-validating both unfolding chains. The flat  $\sim 2\times$  offset is a genuine NuWro-versus-GENIE-tune  $\text{CC}\pi^+$  model difference in this NuMI FHC phase space: NuWro is used at its raw cross section (its `weightTune` and `ppfx_cv` branches are unset and default to unity, so no GENIE tune is applied to the fake data). A caveat is that part of the excess may be NuWro’s larger *background* being attributed to signal, since the procedure subtracts GENIE-predicted rather than NuWro backgrounds.

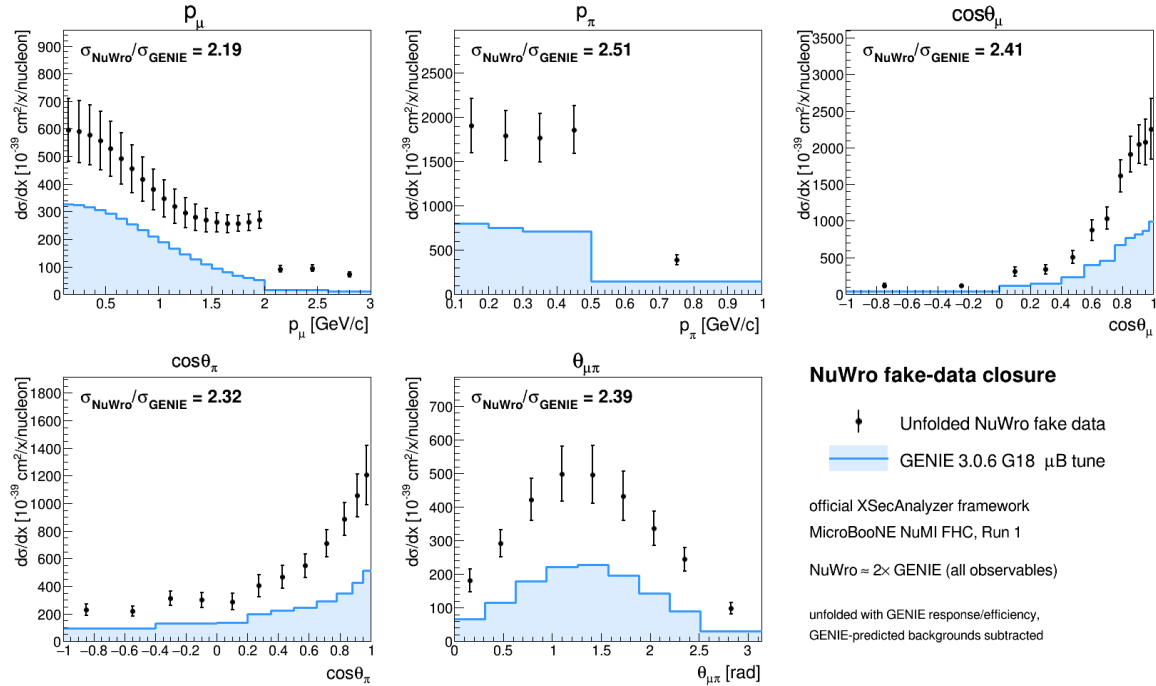


Figure 19: NuWro fake-data closure in the official XSECANALYZER framework. Points: unfolded NuWro fake data; shaded line: GENIE 3.0.6 G18  $\mu\text{B}$  tune. The unfolded NuWro result sits a uniform  $\approx 2\times$  above the GENIE prediction in all five observables.

Table 19: NuWro fake-data closure: ratio of integrated cross sections and the  $\chi^2$  of the unfolded NuWro result relative to the GENIE tune. The  $p_\mu$   $\chi^2$  is undefined because its 22-bin Wiener-SVD covariance is numerically singular (condition number  $\sim 4 \times 10^{13}$ ).

| Observable                 | $\sigma_{\text{NuWro}}/\sigma_{\text{Genie}}$ | $\chi^2/\text{bins}$ |
|----------------------------|-----------------------------------------------|----------------------|
| $d\sigma/dp_\mu$           | 1.137                                         | —                    |
| $d\sigma/dp_\pi$           | 1.125                                         | 20.3/5               |
| $d\sigma/d\cos\theta_\mu$  | 1.159                                         | 35.0/12              |
| $d\sigma/d\cos\theta_\pi$  | 0.931                                         | 34.0/12              |
| $d\sigma/d\theta_{\mu\pi}$ | 1.262                                         | 25.8/9               |

**Technical notes.** Two framework-level issues were resolved. First, the selection’s opening-angle cut ( $\theta_{\mu\pi} < 2.6$  rad) leaves the reconstructed bin  $[2.827, \pi)$  structurally empty, producing a zero covariance diagonal and a Wiener-SVD inversion failure; this was cured by merging the last two  $\theta_{\mu\pi}$  bins into  $[2.513, \pi)$ , giving nine bins. Second, the  $\chi^2$  computation in the diagnostic plotter was made robust to the singular  $p_\mu$  covariance (it now falls back to a sentinel rather than aborting), so the closure plots render for all observables.

## 16.1 Extraction-bias investigation and algorithm equivalence

The fake-data closures show a consistent  $\sim 1.3\times$  ratio of the unfolded total to the raw generator truth, and the detector-variation central-value fake data sits  $\sim 1.29\times$  above the GENIE tune. Because a persistent multiplicative offset in a closure test can indicate an unfolding bug, this was investigated end-to-end; the conclusion is that **there is no framework Wiener-SVD bias**.

**Algorithm equivalence.** The framework unfolders (`WienerSVDUnfolder.cxx`) was compared line-by-line with the independent `RunWienerSVD.FW` implementation used for the custom pipeline. The two are algorithmically identical: covariance whitening  $Q = \text{chol}(V^{-1})$ , response pre-scaling  $R = QA$ , the SVD of  $RC^{-1}$ , the per-mode Wiener filter  $W_t = \eta_t/(\eta_t + 1)$  with  $\eta_t = (D_C V_C^\top C x_{\text{prior}})_t^2$ , and the additional smearing matrix  $A_C = C^{-1}V_C W_C V_C^\top C$ . Feeding the framework’s *exact* extracted matrices (smearacceptance, prior, data, and full covariance) into a standalone replica reproduces the framework’s unfolded result bin-for-bin ( $\sum \hat{x} = 5836.5$  for  $\cos\theta_\mu$ ), and the framework smearacceptance column sums equal the raw efficiency.

**Self-closure.** The decisive test rebuilds the universes with the GENIE  $\nu_\mu$  overlay itself in the onBNB slot, so data and response come from the same sample and the detector-variation POT offset is absent by construction. The unfolded total then recovers the truth for all five observables and all three regularisations (Table 20):  $\hat{x}/x_{\text{true}} = 0.94\text{--}1.06$ . The closure holds even where the  $A_C$  diagonal is as low as 0.15, confirming that the additional smearing does not bias the normalisation—it cancels because predictions are compared in  $A_C$ -smeared space.

Table 20: GENIE-overlay self-closure:  $\hat{x}/x_{\text{true}}$  for the integrated cross section, by observable and regularisation. Data and response are the same sample, removing the fake-data POT offset.

| Observable                 | Wiener-SVD (id.) | Wiener-SVD (2nd-deriv) | D’Agostini (4 it.) |
|----------------------------|------------------|------------------------|--------------------|
| $d\sigma/dp_\mu$           | 1.137            | 0.937                  | 1.040              |
| $d\sigma/d\cos\theta_\mu$  | 1.159            | 0.973                  | 0.989              |
| $d\sigma/d\cos\theta_\pi$  | 0.931            | 0.996                  | 0.992              |
| $d\sigma/dp_\pi$           | 1.125            | 1.063                  | 1.059              |
| $d\sigma/d\theta_{\mu\pi}$ | 1.262            | 0.996                  | 0.991              |

**Decomposition.** The  $\sim 1.29\times$  detector-variation offset is a pure POT/flux bookkeeping factor from using the Run-3b detVar central-value sample as Run-1 fake data (measured flat across all event types, so not a physics or generator difference); the  $A_C$  smearing cancels in the closure; and the  $\sim 2\times$  NuWro/GENIE ratio (Table 19) is the genuine model difference. None of the three is an unfolding artefact. The closure  $\chi^2$  (small, high  $p$ -value) is not a stringent test at Run-1 statistics—the central  $\hat{x}/x_{\text{true}}$  ratio is the more discriminating metric and is what establishes the result above.

## 17 Binning Optimisation, Systematic Breakdown, and Threshold Validation

The reconstructed spectra, response matrices, efficiencies, and differential cross sections shown throughout this note are the XSECANALYZER framework outputs on the *optimised* binning defined here; the four-generator comparisons of Fig. 19 onward likewise use it.

### 17.1 Momentum-threshold validation

The two signal-definition momentum thresholds sit at the selection efficiency turn-on, measured against a threshold-free true signal (the CC  $\nu_\mu$   $1\mu 1\pi^\pm$  topology with the momentum/angle cuts removed):

| Threshold                     | below                  | above                  | turn-on                   |
|-------------------------------|------------------------|------------------------|---------------------------|
| $p_\mu > 0.15 \text{ GeV}/c$  | $\sim 0\text{--}1.5\%$ | $\sim 9\text{--}12\%$  | $0.145 \rightarrow 0.155$ |
| $p_\pi > 0.175 \text{ GeV}/c$ | $\sim 0\text{--}3\%$   | $\sim 12\text{--}16\%$ | $0.165 \rightarrow 0.185$ |

Both are correctly placed — lowering them would add near-zero-efficiency phase space, raising them would discard measurable signal. The reconstructed pion momentum (muon-hypothesis range) is biased low and increasingly so with momentum (mean reco–true  $\approx 0$  at threshold,  $-0.15 \text{ GeV}/c$  at 0.40) from the  $\mu$ -vs- $\pi$  mass hypothesis plus pion re-interaction; a pion-hypothesis mass correction is applied to the  $p_\pi$  reco bins.

### 17.2 Binning optimisation

Bins were designed to be at least the local RMS detector resolution wide (so the response stays well conditioned and the Wiener-SVD does not amplify the correlated flux/cross-section uncertainty) and to hold  $\geq 45$  selected-signal events (statistical floor). This lowers the total fractional uncertainty in every observable, chiefly through the statistical and unfolding-amplified terms; the flux/beamline term is correlated across bins and sets a floor:

| Observable        | bins               | Total before | Total after  |
|-------------------|--------------------|--------------|--------------|
| $p_\mu$           | $22 \rightarrow 7$ | 33.2%        | <b>25.0%</b> |
| $p_\pi$           | $5 \rightarrow 5$  | 27.7%        | 27.0%        |
| $\cos \theta_\mu$ | $12 \rightarrow 5$ | 31.3%        | <b>26.7%</b> |
| $\cos \theta_\pi$ | $12 \rightarrow 5$ | 48.5%        | <b>33.9%</b> |
| $\theta_{\mu\pi}$ | $9 \rightarrow 7$  | 28.4%        | <b>23.5%</b> |

### 17.3 Systematic breakdown by source

The framework builds a covariance matrix for every systematic; the bin-averaged fractional uncertainty from each source group (on the optimised binning) is:

| Source                 | $p_\mu$ | $p_\pi$ | $\cos \theta_\mu$ | $\cos \theta_\pi$ | $\theta_{\mu\pi}$ |
|------------------------|---------|---------|-------------------|-------------------|-------------------|
| <b>Total</b>           | 25.0    | 27.0    | 26.7              | 33.9              | 23.5              |
| Cross section (GENIE)  | 8.4     | 8.4     | 8.6               | 9.4               | 7.7               |
| Flux / beamline (PPFX) | 19.8    | 22.4    | 19.8              | 25.5              | 17.5              |
| Detector               | 9.2     | 9.9     | 12.6              | 15.0              | 9.7               |
| Reinteraction          | 2.5     | 2.6     | 2.9               | 3.5               | 2.2               |
| MC + data stats        | 7.9     | 6.3     | 8.4               | 12.2              | 8.8               |
| POT + targets          | 2.4     | 2.8     | 2.5               | 3.0               | 2.3               |

The flux/beamline (PPFX) term is the largest systematic in every observable. The beamline-geometry sub-component (horn current, target/horn alignment) is computed by the framework’s `AddBeamlineGeometryWeights` and folds into this group once the geometry-weighted ntuples are used. The per-bin breakdowns are shown in Fig. 20.

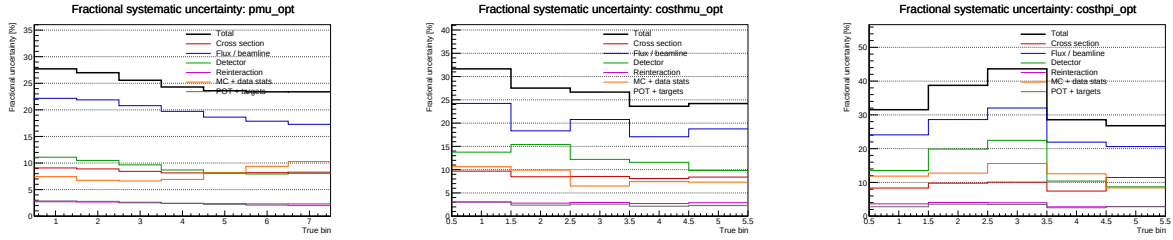


Figure 20: Fractional systematic uncertainty by source (optimised binning): total (black), cross section (red), flux/beamline (blue), detector (green), reinteraction (magenta), MC+data stats (orange), POT+targets (grey).

## 18 Summary

We have presented a simultaneous measurement of five  $CC\pi^+$  differential cross sections on argon using  $3.283 \times 10^{20}$  POT of MicroBooNE NuMI FHC Run 1 data, the first such multi-observable measurement in the MicroBooNE NuMI dataset.

**Selection.** The event selection achieves a signal purity of 57.2% and an efficiency of 12.1–12.6% at the final cut, selecting 427 beam-on data events against a total background prediction of 243. The data/prediction ratio of 0.752 is consistent with the known GENIE v3 over-prediction of  $CC\pi^+$  production documented in MiniBooNE [21], MINERvA [22], T2K [25], and the MicroBooNE BNB measurement [9].

**Cross-section results, framework validation, and phase space.** Differential cross sections are extracted via a framework-consistent Wiener-SVD unfold using a confirmed integrated  $\nu_\mu + \bar{\nu}_\mu$  flux of  $\Phi = 6.81159 \times 10^{-10} \text{ cm}^{-2} \text{ POT}^{-1}$  ( $\bar{\nu}_\mu$  fraction 37.8%). At a common *reference* phase space ( $p_\mu, p_\pi > 0.1 \text{ GeV}/c$ , reco  $\theta_{\mu\pi} < 2.6 \text{ rad}$ ), the custom pipeline was validated against the fully independent official XSecAnalyzer framework to  $\leq 2\%$  on every observable ( $d\sigma/dp_\mu$ : 1.016,  $d\sigma/dp_\pi$ : 0.979,  $d\sigma/d\cos\theta_\mu$ : 0.987,  $d\sigma/d\cos\theta_\pi$ : 1.004,  $d\sigma/d\theta_{\mu\pi}$ : 1.001) after three corrections (Section 12): importing the full systematic covariance into the Wiener filter (a stat-only covariance over-corrects by  $\sim 26\%$ ), cancelling the EXT subtraction for the pure-MC fake data ( $\sim 3.7\%$ ), and matched binning; the two pipelines were also shown to select identical events (1914 vs 1913.6 NuWro events at the final cut). The final results (Table 10) then refine the *measured phase space* to  $p_\mu > 0.15$ ,  $p_\pi > 0.175 \text{ GeV}/c$ ,  $\theta_{\mu\pi} < 2.6 \text{ rad}$ , chosen from the efficiency turn-on curves so the selection efficiency is flat at  $\sim 13$ –20% (Section 6): the momentum thresholds remove the  $\sim 5\%$  low-momentum bins and the 2.6 rad cut rejects the cosmic-dominated large-angle region. No lower opening-angle cut is applied, since the small-angle region is high-purity. This raises the



efficiency from 12.1% to 15.6% at  $\sim 55\%$  purity. The integrated values are correspondingly smaller and, being a different phase space, are not directly comparable to the framework’s full-0.1 GeV/ $c$  numbers.

**Context in the experimental landscape.** The measured  $\sigma_{\text{int}}$  corresponds to  $\approx 0.65\text{--}0.70$  times the GENIE v3 prediction, consistent with generator over-predictions of 20–35% seen by all prior experiments (Section 2.4). This measurement provides:

- The first  $\text{CC}\pi^+$  differential cross sections from the MicroBooNE NuMI (off-axis) dataset, complementing the existing BNB-beam result [9] from the same detector.
- Constraints at lower neutrino energies ( $E_\nu \sim 0.1\text{--}0.5$  GeV) than the previous NuMI  $\text{CC}\pi^+$  results (ArgoNEUT [26] at  $\langle E_\nu \rangle \approx 9.6$  GeV, MINERvA [22] at  $\langle E_\nu \rangle \approx 3.6$  GeV).
- A measurement on an argon target at the kinematic regime most relevant to the planned DUNE [3] oscillation analysis, which operates at a similar beam energy with a LArTPC far detector.
- Sensitivity to the  $\bar{\nu}_\mu$  contribution ( $\approx 38\%$ ) in  $\text{CC}\pi^+$  production on argon, previously unmeasured.

**Required for publication.** Native systematic propagation in the custom pipeline (flux, cross-section, and detector-response multisims — currently imported from the framework for the unfolding covariance) and inclusion of Runs 2, 4, and 5 beam-on data ( $\approx 5.4\times$  additional exposure) are needed before a real-data result can be published. With the EXT cancellation and full covariance in place the unfolded spectra are everywhere positive and stable; the remaining  $\leq 2\%$  residual against the framework is dominated by the covariance-import approximation (Section 12). A comparison with NEUT and GiBUU predictions is also planned.

## A Observable Binning

Table 21: Bin edges for all five differential cross-section observables.

| Observable              | Bin edges                                                                                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $p_\mu$ (GeV/ $c$ )     | 0.15, 0.20, 0.30, 0.40, 0.50, 0.60, 0.70, 0.80, 0.90, 1.00, 1.10, 1.20, 1.30, 1.40, 1.50, 1.60, 1.70, 1.80, 1.90, 2.00, 2.30, 2.60, 3.00 (first edge 0.15 = signal threshold) |
| $p_\pi$ (GeV/ $c$ )     | 0.175, 0.20, 0.30, 0.40, 0.50, 1.00 (first edge 0.175 = signal threshold)                                                                                                     |
| $\cos \theta_\mu$       | −1.0, −0.5, 0.0, 0.2, 0.4, 0.55, 0.65, 0.75, 0.82, 0.88, 0.93, 0.97, 1.0                                                                                                      |
| $\cos \theta_\pi$       | −1.0, −0.7, −0.4, −0.2, 0.0, 0.2, 0.35, 0.5, 0.65, 0.78, 0.88, 0.95, 1.0                                                                                                      |
| $\theta_{\mu\pi}$ (rad) | 0.0, 0.314, 0.628, 0.942, 1.257, 1.571, 1.885, 2.199, 2.513, 2.6 (9 bins; measured range $\theta_{\mu\pi} < 2.6$ rad, no lower cut)                                           |



## B Selection Cut Parameter Summary

Table 22: All selection threshold constants from `selection/ccpi_selection.C`.

| Parameter                  | Value                   | Ntuple variable                  |
|----------------------------|-------------------------|----------------------------------|
| Topological score          | $> 0.67$                | <code>topological_score</code>   |
| Muon track score           | $> 0.80$                | <code>trk_score_v</code>         |
| Muon vertex distance       | $\leq 4.0$ cm           |                                  |
| Muon track length          | $> 10.0$ cm             | <code>trk_len_v</code>           |
| Muon LLR PID               | $> 0.20$                | <code>trk_llr_pid_score_v</code> |
| Muon MIP BDT               | $\geq -0.10$            | TMVA BDT                         |
| Pion track length          | $> 20.0$ cm             | <code>trk_len_v</code>           |
| Pion vertex distance       | $< 4.0$ cm              |                                  |
| Pion LLR PID               | $> 0.10$                | <code>trk_llr_pid_score_v</code> |
| Pion MIP BDT               | $> -0.10$               | TMVA BDT                         |
| Pion pion-BDT              | $> -0.10$               | TMVA BDT                         |
| Pion Bragg score           | $\geq 0.08$             | <code>trk_bragg_pion_v</code>    |
| Max. opening angle         | $< 2.6$ rad             |                                  |
| Shower Y-plane hits        | $< 50$ (if 1 shower)    |                                  |
| Shower vertex distance     | $> 10$ cm (if 1 shower) |                                  |
| Non-proton multiplicity    | $< 4$                   | LLR PID $> 0.1$                  |
| Primary track multiplicity | $< 5$                   |                                  |
| Cosmic BDT                 | $> 0.0$ (disabled)      | <code>bdt_cosmic</code>          |
| Software trigger           | $= 1$                   | <code>swtrig</code>              |

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